SUDOSCAN

User guide

Instruction for use

Version 1.5



impetomedical

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Introduction

Use

Principle

SUDOSCAN measures the capacity of the sweat glands to release chlorides ions in response to an electrochemical activation. It is a dynamic test equivalent to a stress test. The information is then used to determine peripheral sudomotor dysfunction.

Description

SUDOSCAN is an electro-medical device that consists of:



- **A control panel and display**
- **A pair of foot electrodes**
- **A pair of hand electrodes**
- **A pair of head electrodes**
- **A power cable**

SUDOSCAN is powered by Windows XP and/or Windows7.

Installation

Carefully follow instructions in this manual to ensure correct installation.

Complete installation should take no more than 15-20 minutes.

1. Unpacking

SUDOSCAN is shipped in a single transportation box with an Impeto Medical logo on the outside. Inside there are 3 smaller boxes:

1. A medium white box containing the reusable feet and hand electrodes,
2. A small white box containing the auxiliary cable and the reusable head electrodes,
3. A big white box, with the SUDOSCAN Master Unit and the power cable.



1.1. Unpack the master unit

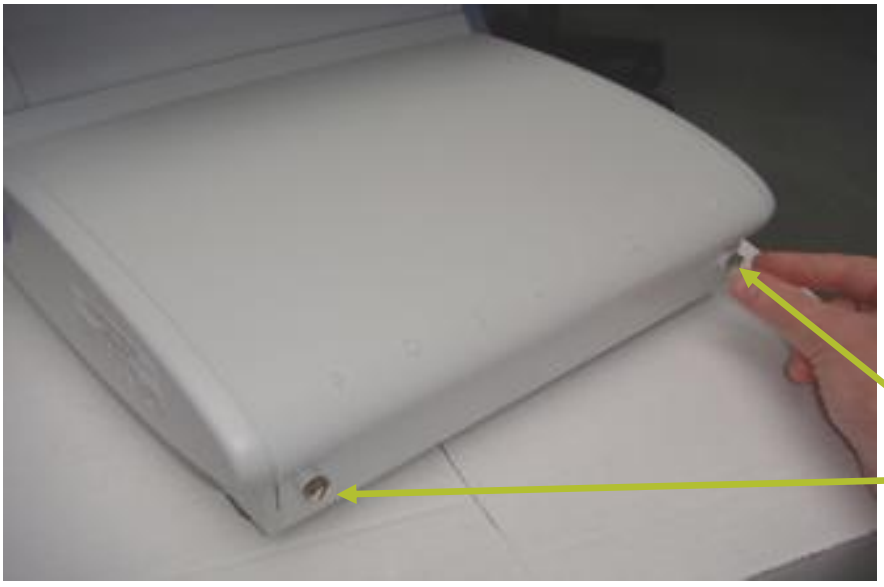
1	Remove the Master Unit along with its 2 molded protection foams from the big white box.
2	Put the Master Unit on one side to remove the first protection easily and safely the first protection foam.
3	Put the Master Unit on its other side and remove the second protection foam, as well as the plastic bag.

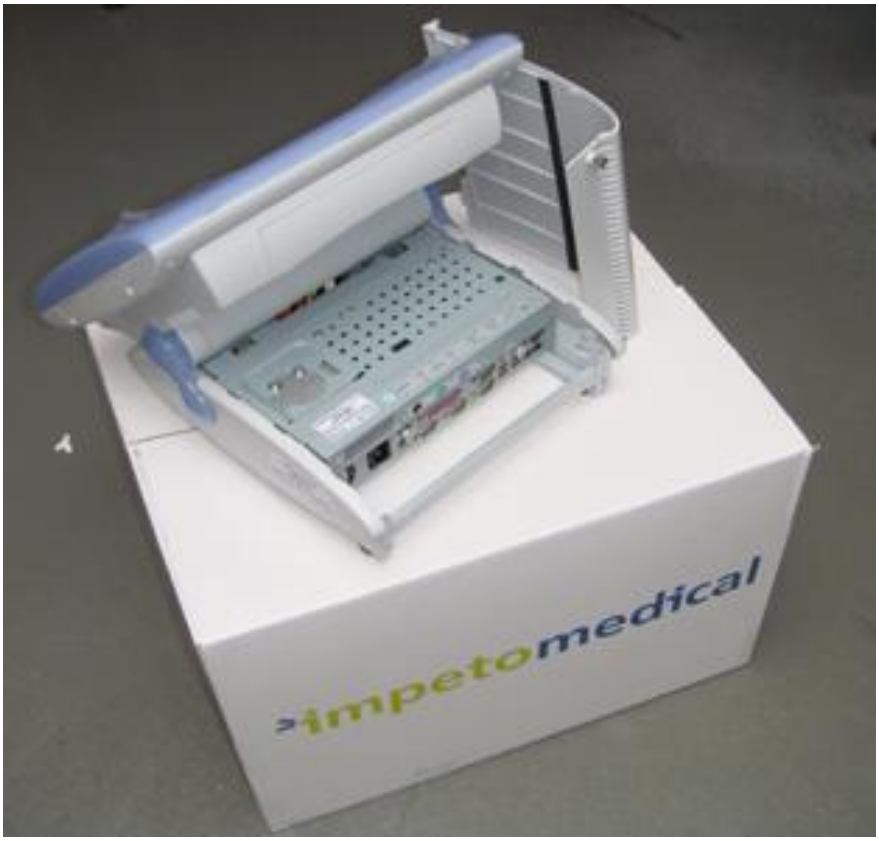




4	Keep the protection foams and plastic bag in the outer box, for future transportation needs.
5	Loosen the 2 retaining screws and remove the rear cover.


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Retaining
screws


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1.2. Unpack the electrodes

1 Remove the electrodes from the medium white box and separate them from their protective foams.



2 Remove the head electrodes cable and the cleaning spray from the small box.

3 Remove all 3 cables and connectors from their plastic protection bags.


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2. Installation

1 Place the Master Unit on a desk or preferably a high table (h > 90 cm/3ft) with the screen facing the user.



NOTE

Please consult your distributor for the availability of an optional SUDOSCAN companion rolling stand which can also conveniently host a printer.

Connecting the electrodes

2 Place the hands electrodes in front of the Master Unit, guide the cable under the metallic bar at the rear, and insert the MiniDin connector at the end of the Hands cable to the 'HANDS' connector (Green) at the back of the Master Unit :



3 Make sure to match the color of both socket and plug.

4 Push until a ‘click’ sound indicates the locking of the connector.



5 Double-check the locking by grabbing the cable itself gently backwards.



6 Place the Feet electrode on the floor in front of the Master Unit and connect its cable to the 'FEET' connector (Blue) at the back of the Master Unit, using the procedure above.

7 Connect as well the Head electrodes cable to the 'HEAD' connector at the back of the Master Unit using the same procedure.

WARNING

Please cross check and verify that all electrode cables are connected to their respective receptacle (Head to Head, Hands to HANDS and Feet to FEET) and there is no inadvertent swapping.



8 Guide the power cable under the metallic bar at the rear of the Master Unit and insert it into the PC standard 3-plug power socket at the back of the Master Unit.



Power Cable

9 Connect the power cable to the mains power socket on the wall. Double check the power socket can provide proper grounding.

NOTE

The supplied power cable is intended for EU countries.

For countries outside EU, please make sure using a national PC-type power cable with grounding connector. Make sure also correct grounding is provided at the wall socket.

3. Quick test and configuration

3.1. Powering on SUDOSCAN

1 Open the door located below the screen at the front of the Master Unit.
Inside on the right there are 2 USB ports and 1 on/off button.



2 Press and release the on/off button to power up the Unit.
The Master Unit will run a self-check and then automatically display the SUDOSCAN home page.



3.2. Initial checking

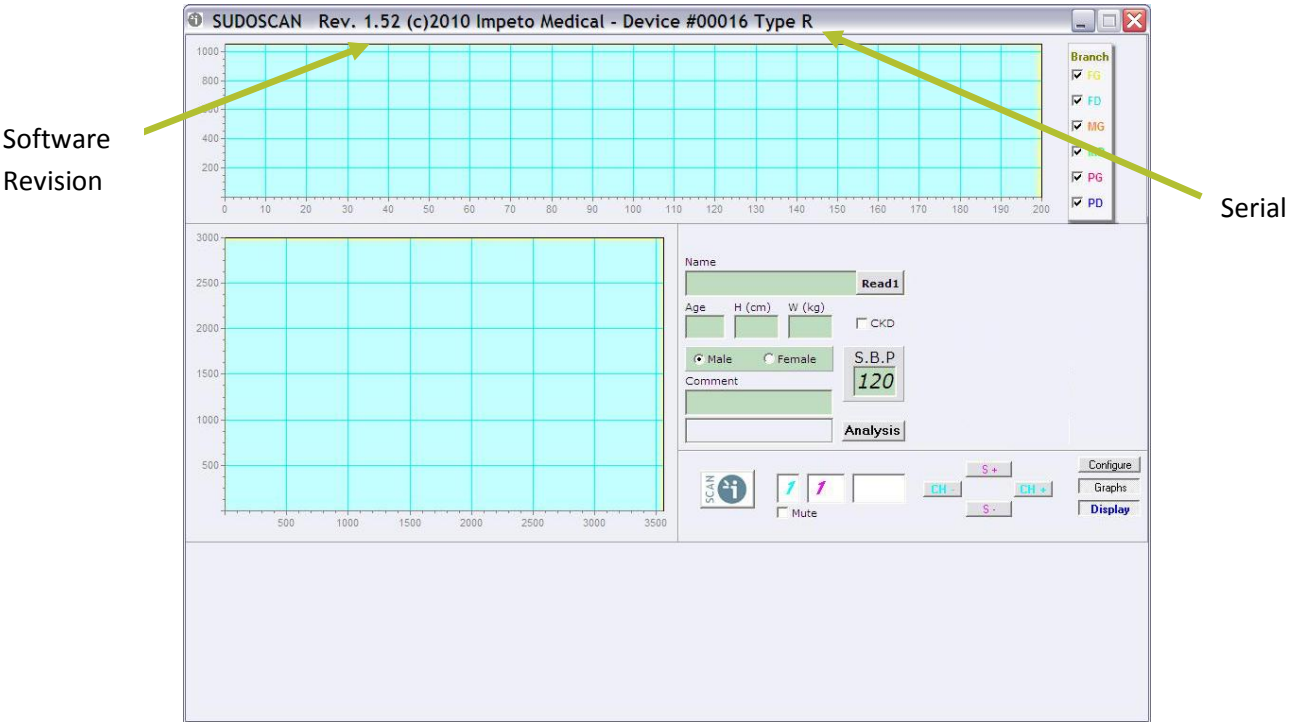


After initialization, the blue LED at the rear of the Master Unit should flash intermittently. The top row on the screen should display:

SUDOSCAN - rev. X.XXx ©2005-2010 Impeto Medical - device #YYYYY type Z

Where:

- X.XXx indicates the software revision number,
- YYYYY indicates the serial number within the product model type Z.



IF: 1/ the blue LED is NOT ON

OR: 2/ the following message appear at the top of the screen:

"SUDOSCAN - rev. X.XXx ©2005-2010 Impeto Medical – not connected",

There is a problem with your SUDOSCAN Master Unit. Please call Impeto Medical or your distributor for assistance.

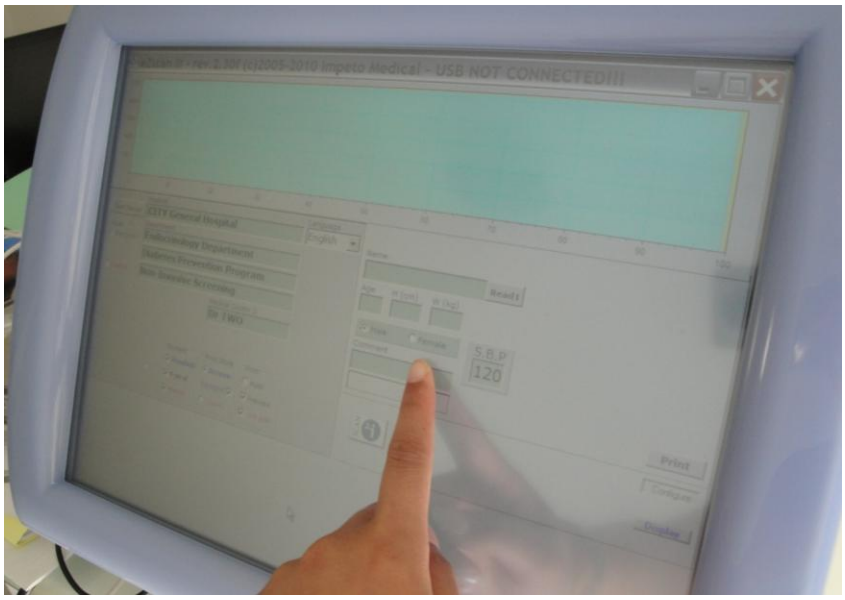
Once this quick check procedure is successfully completed, replace the rear cover at the back of the Master Unit. Secure the cover in place using the 2 retaining screws.



3.3. Selection on the SUDOSCAN tactile screen

SUDOSCAN has a tactile screen. Touch the screen with your finger to make a selection.

You can also use a mouse by plugging one into one of the USB port.

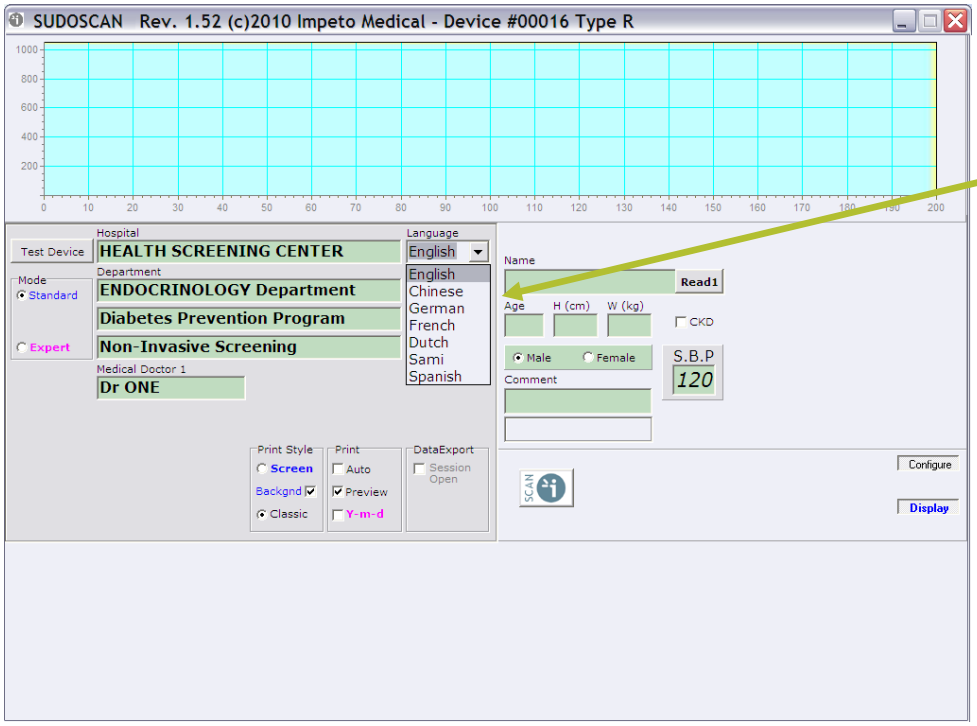


3.4. Language selection

SUDOSCAN software can be used in several languages:

- English
- Chinese
- German
- French
- Dutch
- Spanish
- …

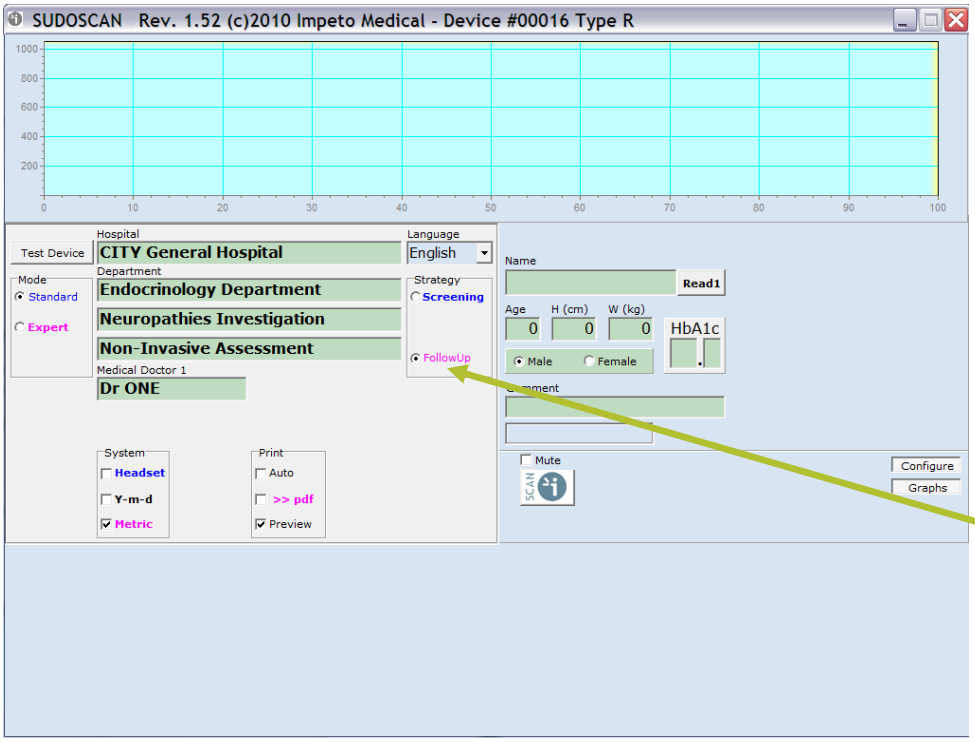
Select one of the languages in the language menu.



3.5. Strategy selection

SUDOSCAN is available through the “Follow-up” strategy.

Check the Follow-up strategy.



The “Screening” strategy enables to evaluate the diabetes risk through EZSCAN.

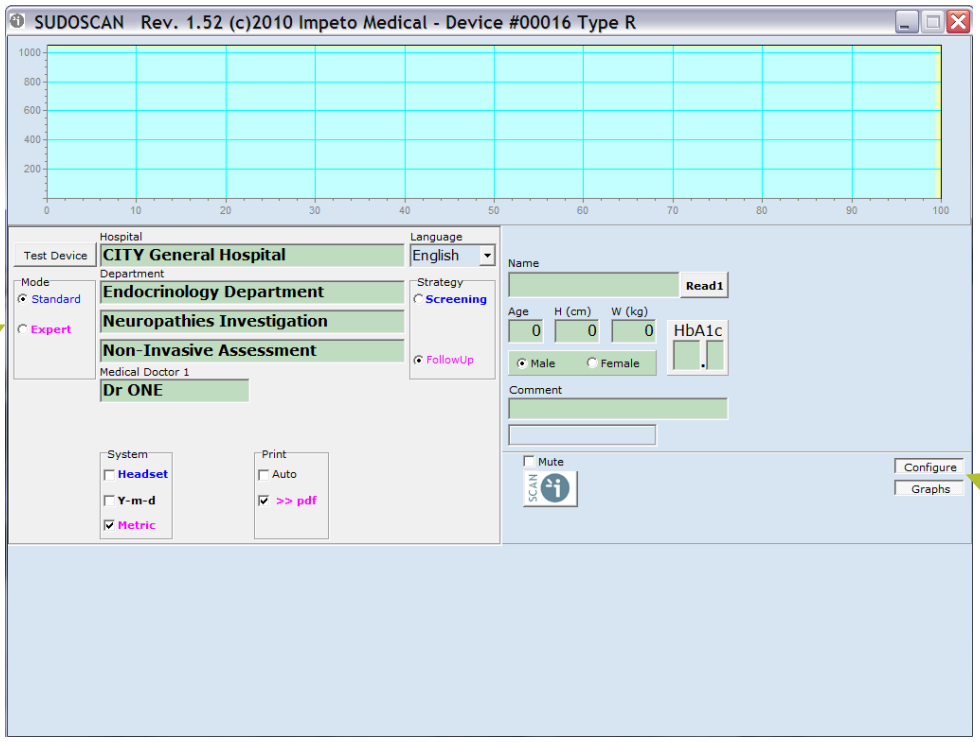
3.6. Mode selection

Two modes are available on SUDOSCAN:

The standard mode is the preferred mode to easily scan a new patient. It is a simple mode showing only the information needed to screen new patients.

The expert mode is the preferred mode to compare with scans previously done.

Check the mode button according to your choice.



Mode selection

Configure button

NOTE

Pressing the “Configure” button on the right will close/open the Configuration window.

3.7. Metric selection

Two measurement systems are available for patients height and weight:

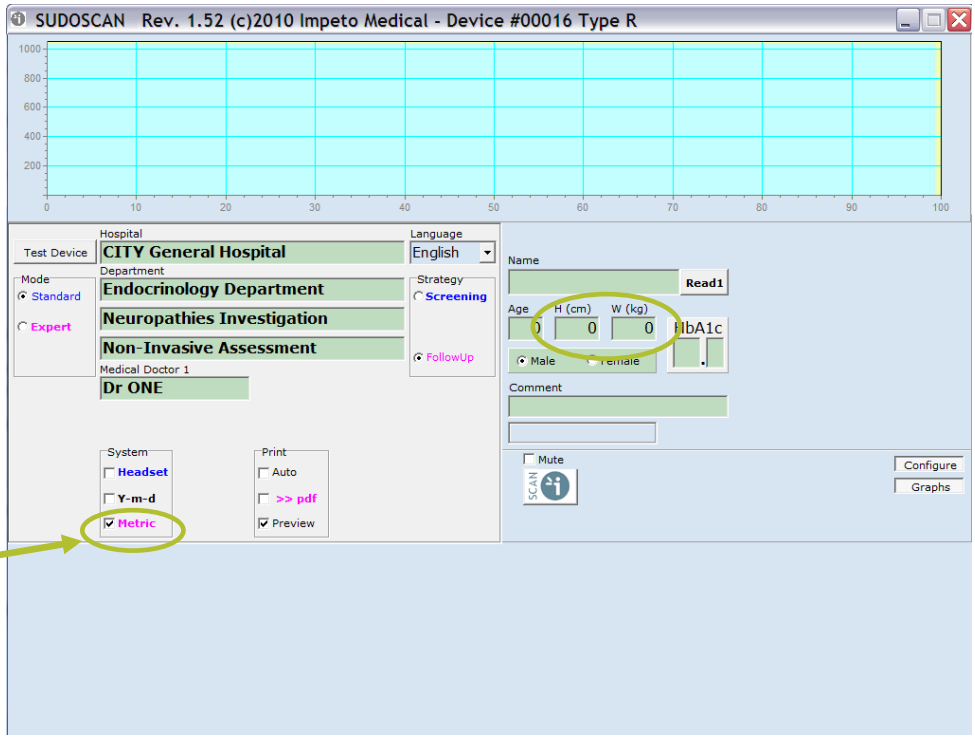
- Metre (cm) and kilogram (kg)
- Feet/inch (feet/in) and Pound (lb)

Check the Metric box if you wish to use cm and kg.

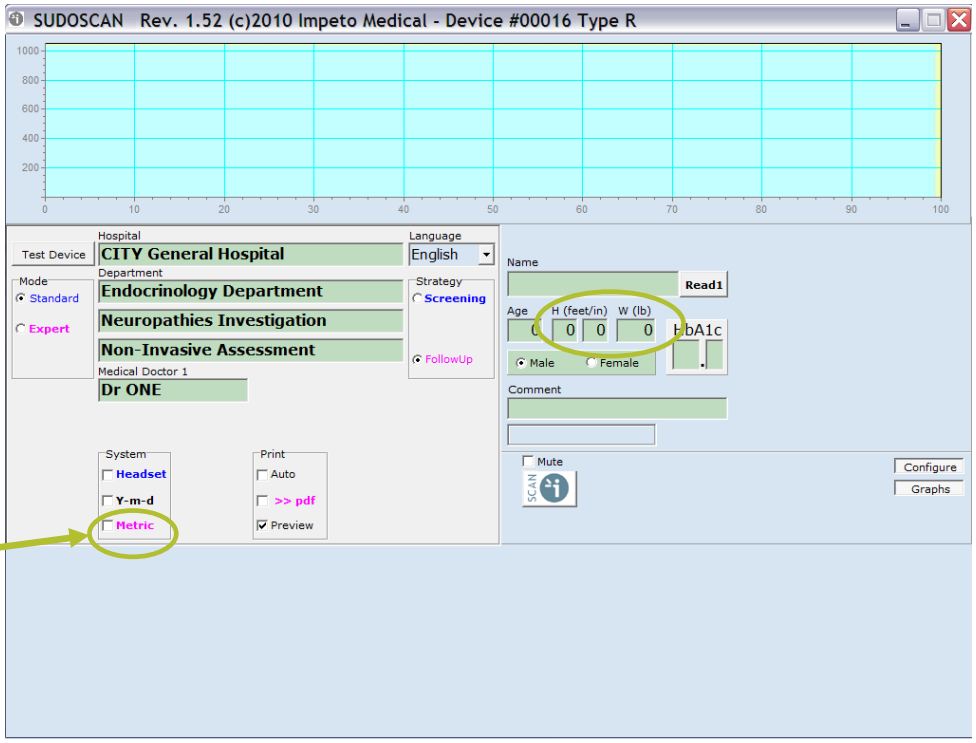
Uncheck the Metric box if you wish to use Feet and Pound.

The measurement will be instantly modified.

Metric checkbox



Metric checkbox



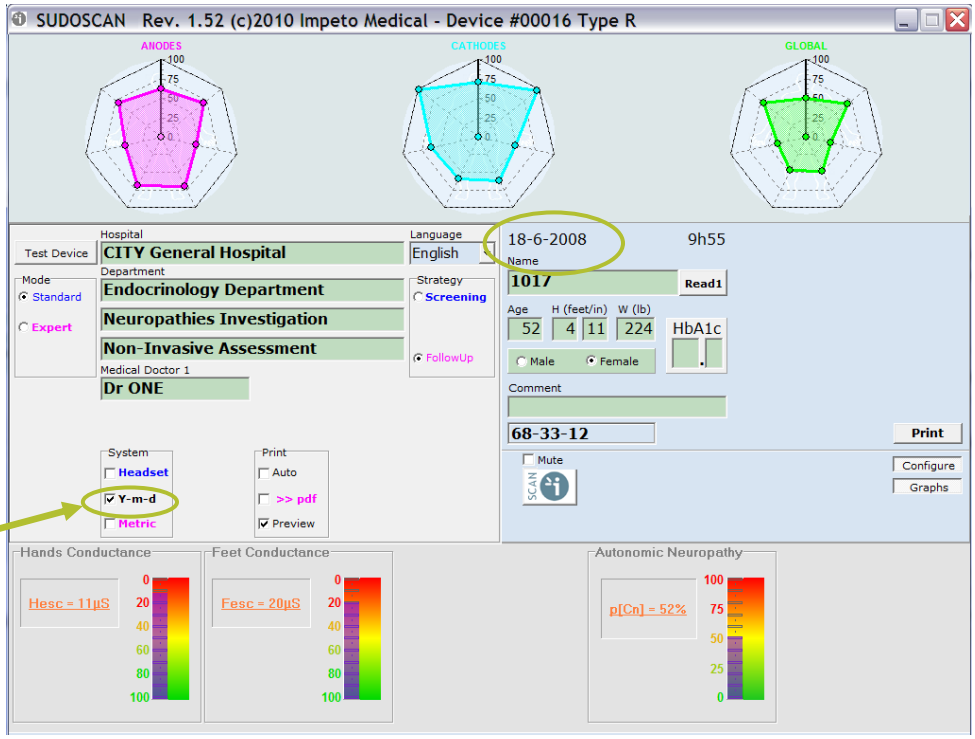
3.8. Date format selection

Two date format are available: Day Month Year (D-m-Y) or Year Month Day (Y-m-d)

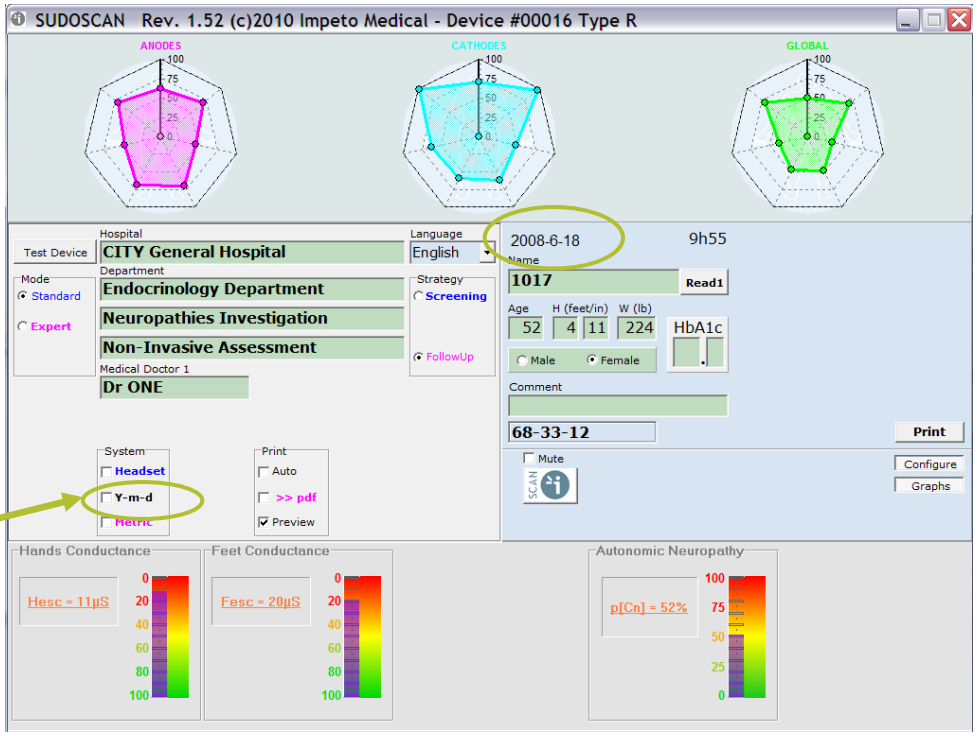
Check the date format checkbox if you wish to use Y-m-d format.

Uncheck the date format checkbox if you wish to use D-m-Y format.

This option will be taken into account for the next scan.



Date format checkbox

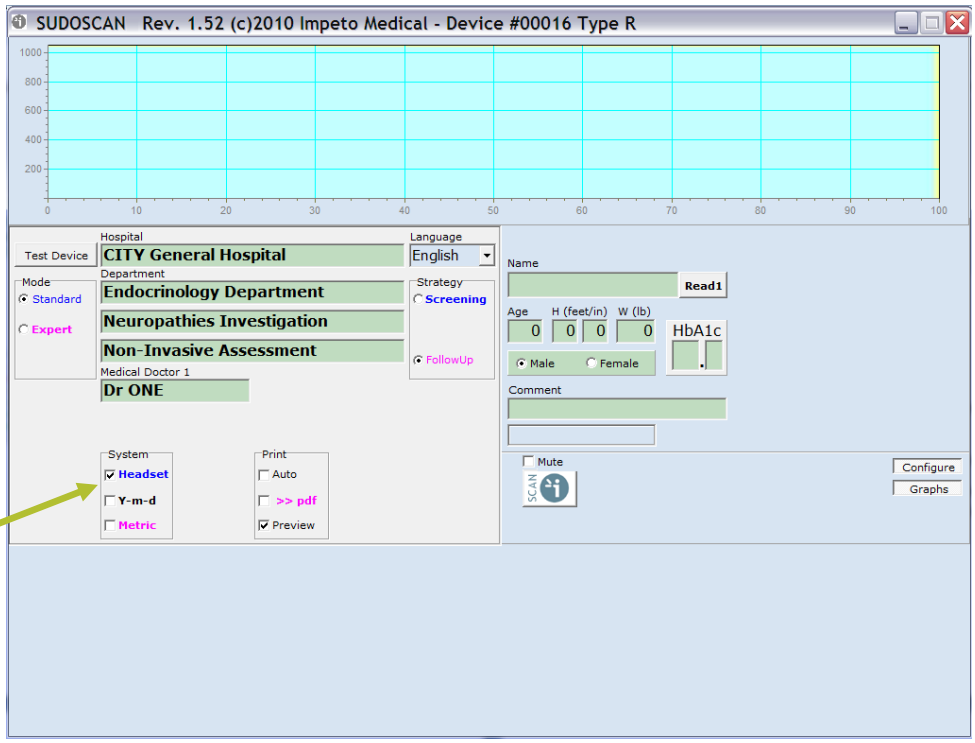


Date format checkbox

3.9. Headset selection

In order to increase the patient throughput on SUDOSCAN device it is possible to remove the Headset electrodes, which will reduce drastically the patient installation time as well as the scanning time.

Uncheck the Headset bow if you wish to use SUDOSCAN without the head electrodes.



Headset checkbox

NOTES

After deciding not to use Headset electrodes, please double check this option is unchecked, in case you experience a “Please check Headset…” error message at the beginning of a new scan.

4. Standard Mode

The standard mode is used to easily scan a new patient. More options are available in the Expert mode (cf. section 5). Check the options selected in the Expert mode before starting a run.

4.1. Quick run

Before starting a scan, you will need the patient to stand barefoot on the Feet electrodes and input a few patient details (name, age, height, weight, gender) using the touch screen display and the virtual keyboard appearing on the screen.

Inform the patient that this test is non-invasive and will take less than 3 minutes to perform.

NOTE

For conventional data input, it is also possible to use any standard PC-type keyboard connected to one of the 6 USB connectors available on Master Unit's front or rear panels.

4.1.1. Patient set-up

1. Ask the patient to remove their shoes and socks.
2. Patient must stand, with bare feet right in the middle of the Feet electrodes.
3. Input patient data using the touch screen virtual keyboard.

   Enter each data field by firmly double-clicking on the corresponding green zones, which will make the appropriate virtual keyboard appear.
   The TAB key on the keyboard can also be used to jump from one field to the next.

   Please key in the correct information for each new patient:

   - *NAME or Patient ID (IDENTITY CODE)*
   - *AGE*
   - *HEIGHT*
   - *WEIGHT*
   - *Select the appropriate GENDER*

- *Enter comments in the "comment" field if necessary.*

There is NO NEED to press the ENTER key after each field entry.

NOTE

Double-check the data with your patient. An invalid data will be notified with a RED background and should prompt entering a valid data.

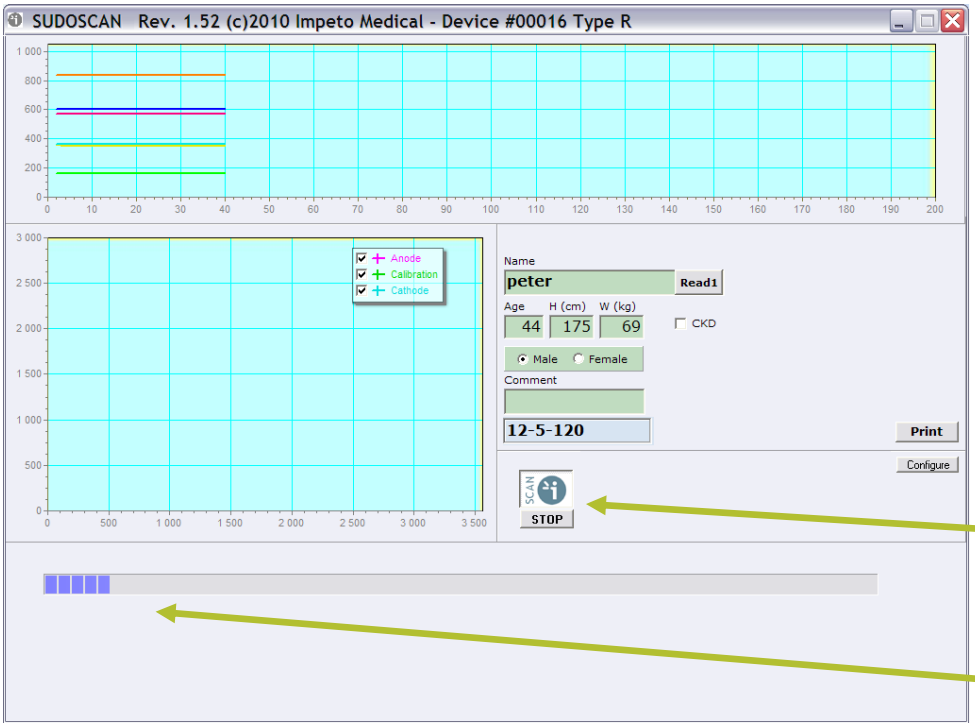
4

Ask the patient to put his hands flat in the middle of the Hands electrodes.
Hands must be applied firmly.

4.1.2. Start the measurement

Press the “SCAN” button to start the scanning procedure.

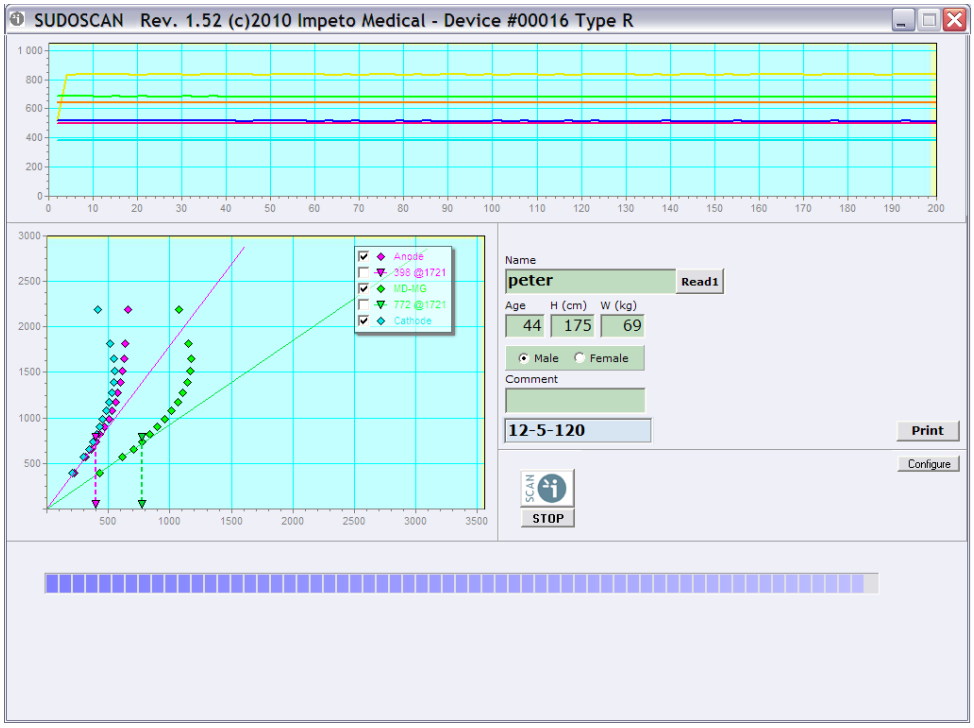
During scanning, the patient is required not to move any parts of the hands or feet in contact with electrode surfaces.



SCAN button

Progress indicator

The **patient should keep still** until the progress indicator shows the scan is completed.



Scan progress can be monitored from the indicator at the bottom of the display.

Scan completion is confirmed by a double-bell sound.

4.1.3. Save patient data

Upon completion of the scan, results are automatically displayed and all patients' data are automatically saved onto the Master Unit's hard disk.

4.2. Clean the electrodes

IMPORTANT

Immediately after each scan, it is important to clean all electrodes with the proper cleaning spray and soft tissues*.

It is not only necessary for health reasons, but it will also 'neutralize' the electrochemical reactions which have taken place on the electrodes during the scan, stopping the corrosion process which would otherwise quickly damage the electrodes surface.

1. Deposit only a small quantity of cleaning product on each electrode.

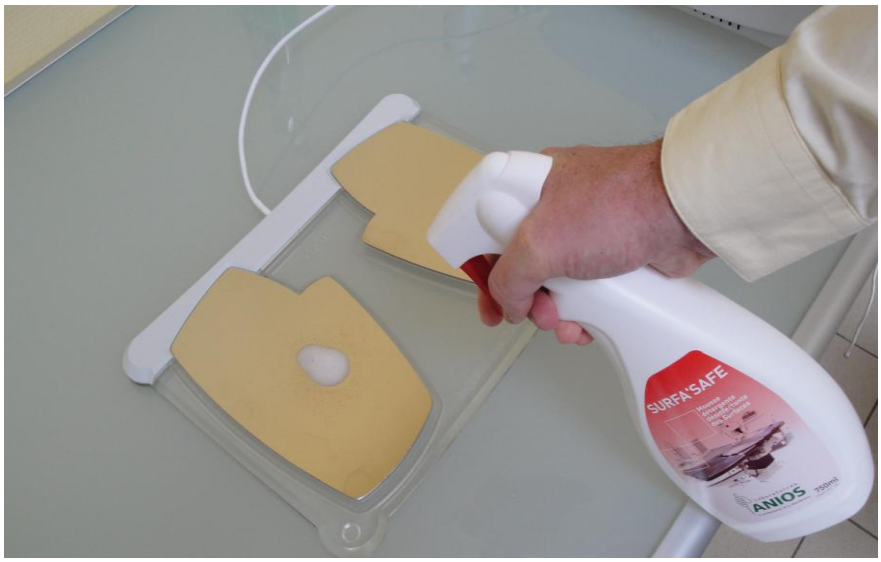
   Avoid applying excessive cleaning liquid.

2. Wipe electrodes with a sheet of soft paper towel to until dry.

   Use a new sheet for each electrode or as long as the surface of the electrodes are not completely dry.

HINT

Take advantage of the shinny mirror aspect of the electrodes to remove all traces from all areas!



***Impeto Medical recommends using Surfa'Safe from ANIOS Laboratories (France) or equivalent.**

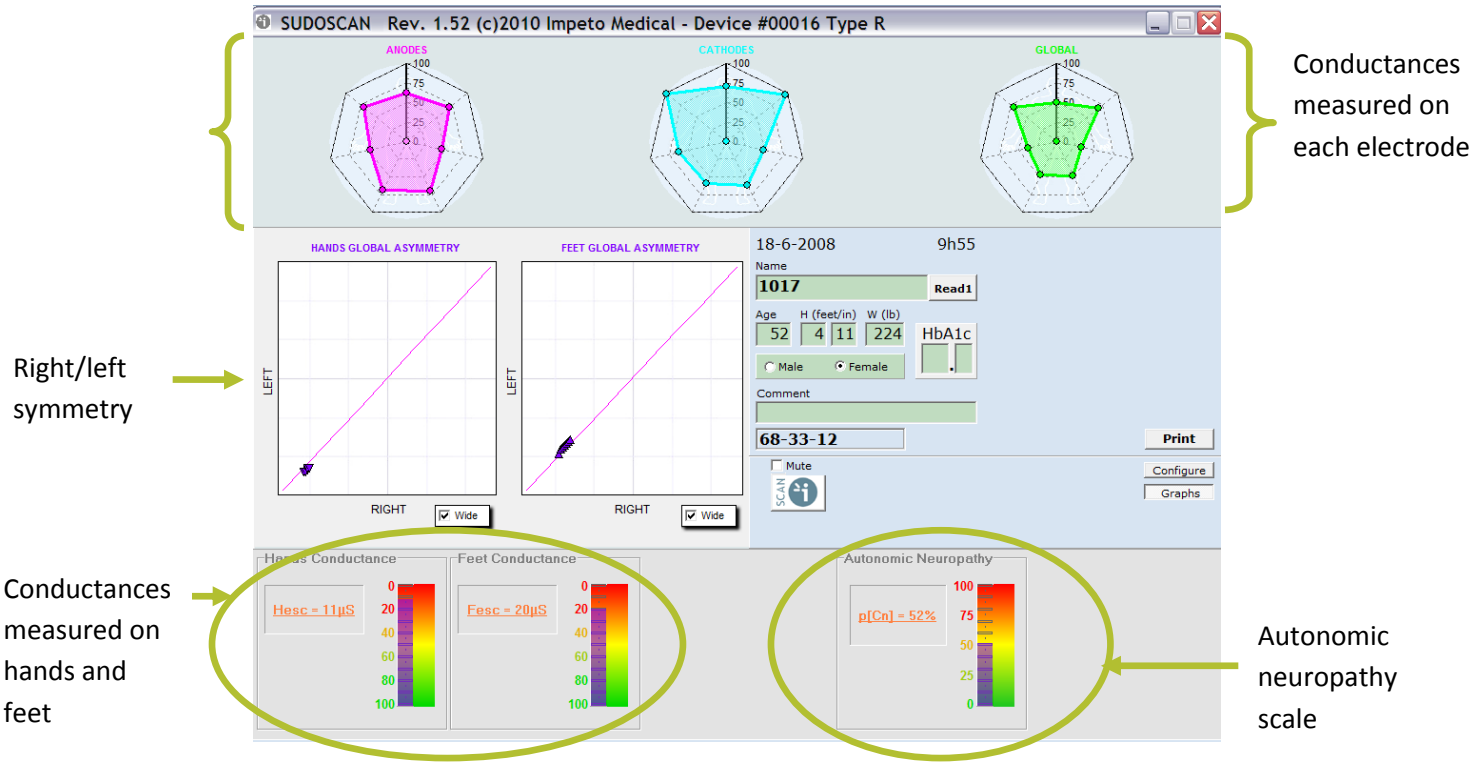
One 750ml bottle will allow to clean all electrodes after each of 750 patients.

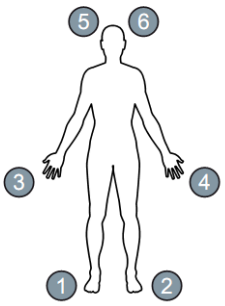
4.3. The SUDOSCAN results

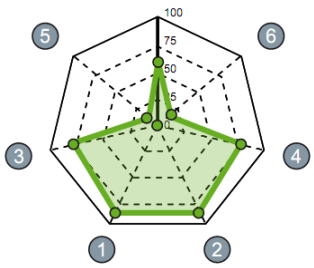
Results should be interpreted by a medical professional and take into account the patient's clinical history.

4.3.1. The screen results

Following a three minute data sampling period, SUDOSCAN allows simple reading of results. The measured conductances are displayed on the top of the screen.

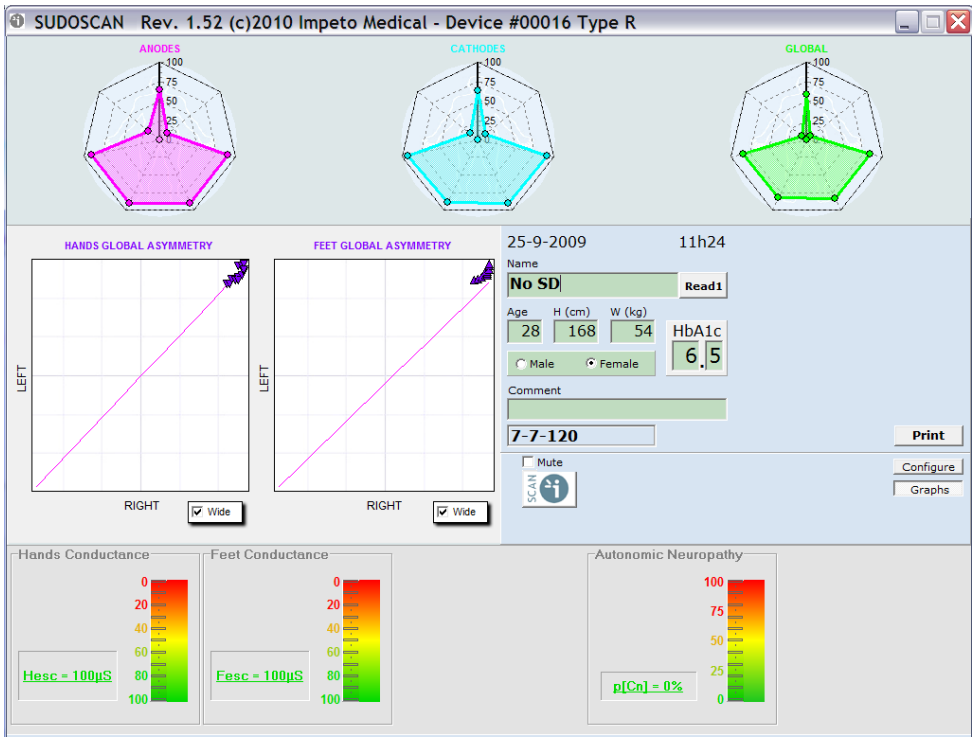






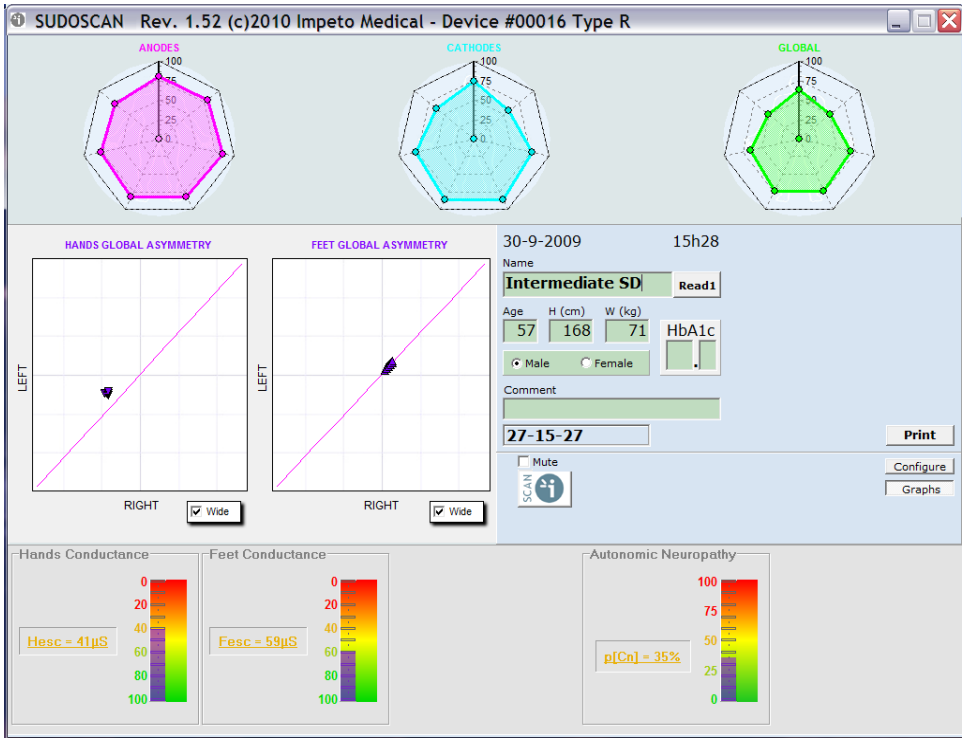
4.3.2. Quick interpretation of the results

4.3.2.1. Patient with no sudomotor dysfunction



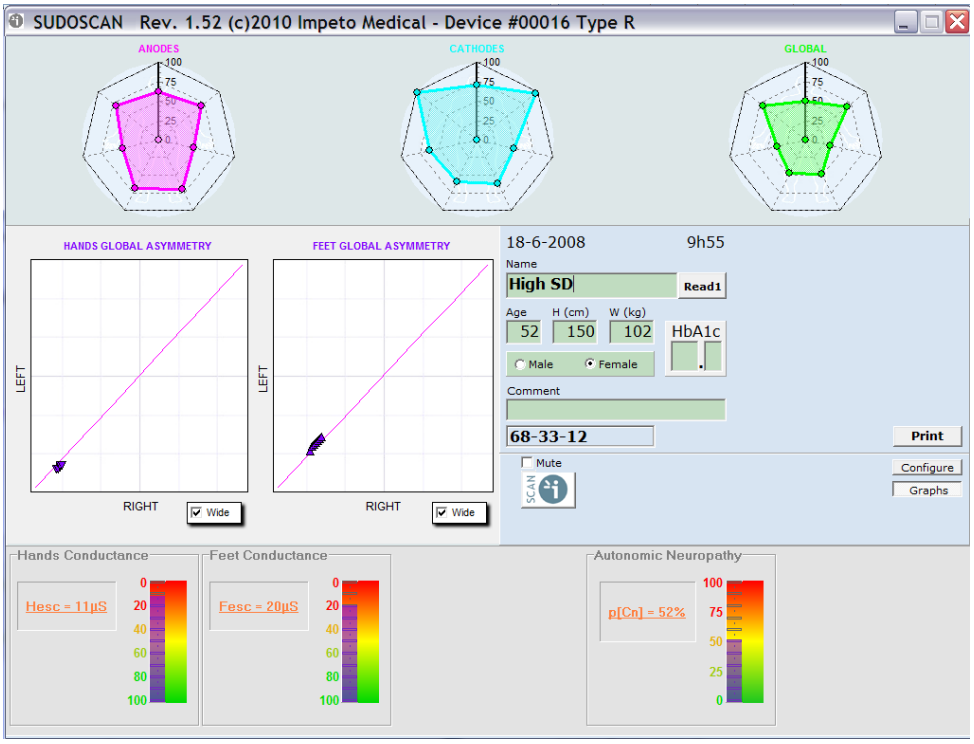
The subject has high conductance on hands and feet (>60µS) which suggest no sudomotor dysfunction.

4.3.2.2. Patient with intermediate sudomotor dysfunctions



The patient has intermediate conductance on hands and feet comprised between [40 – 60] µS. That suggests moderate sudomotor dysfunctions on hands and feet.

4.3.2.3. Patient with intermediate high sudomotor dysfunctions

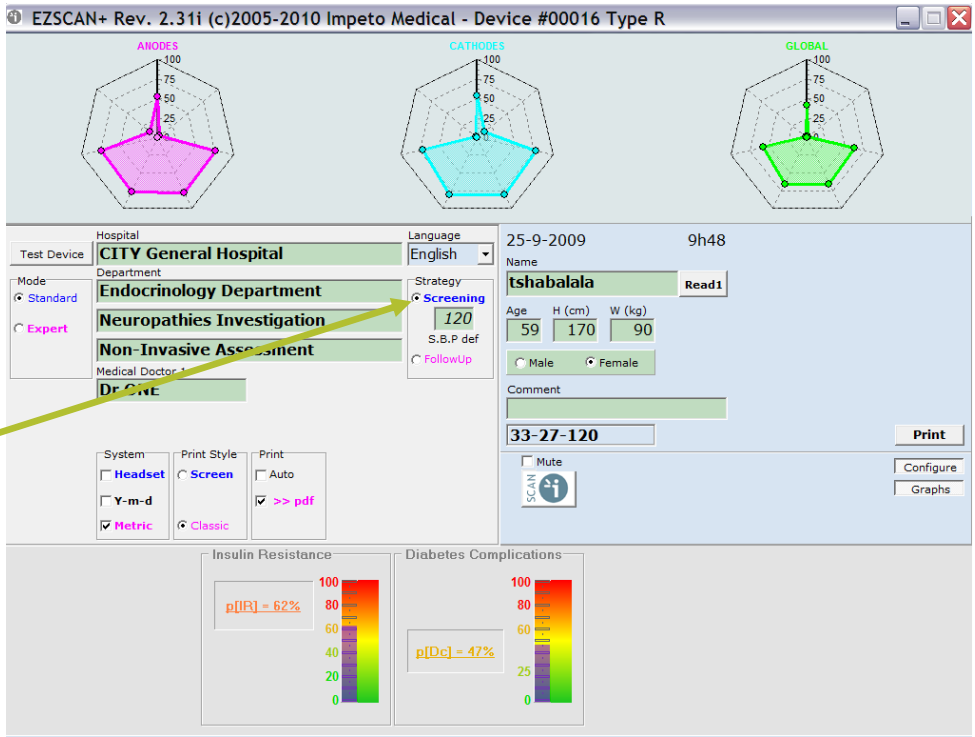


The patient has very low sweat conductances on hands and feet (<40µS). This suggests high sudomotor dysfunctions on hands and feet.

4.4. The EZSCAN results

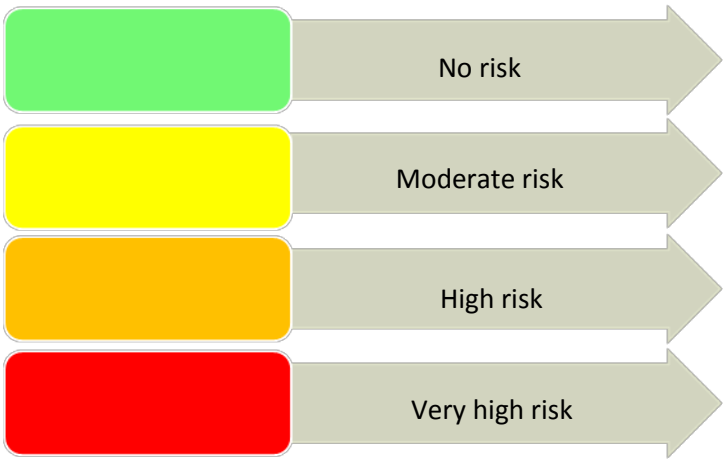
1	Press the configure button.
2	Check the “screening” strategy.
3	Press the configure button.

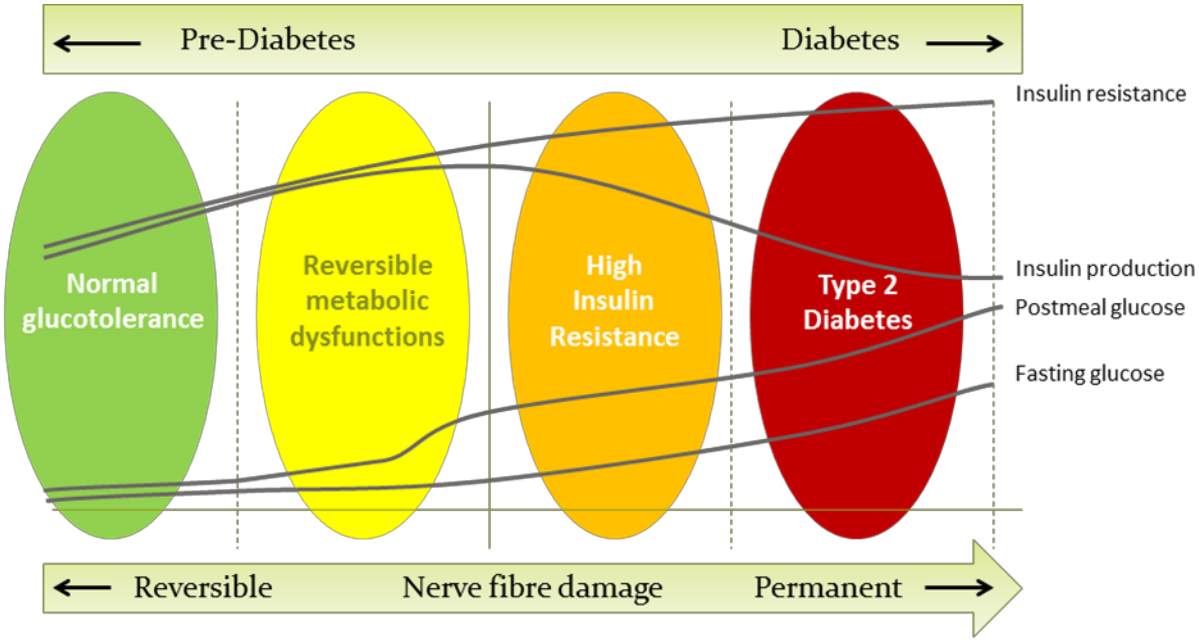
The screening strategy allows simple reading of results with the colored-ellipse positions which indicate where the patient is on the diabetes historical curves’.



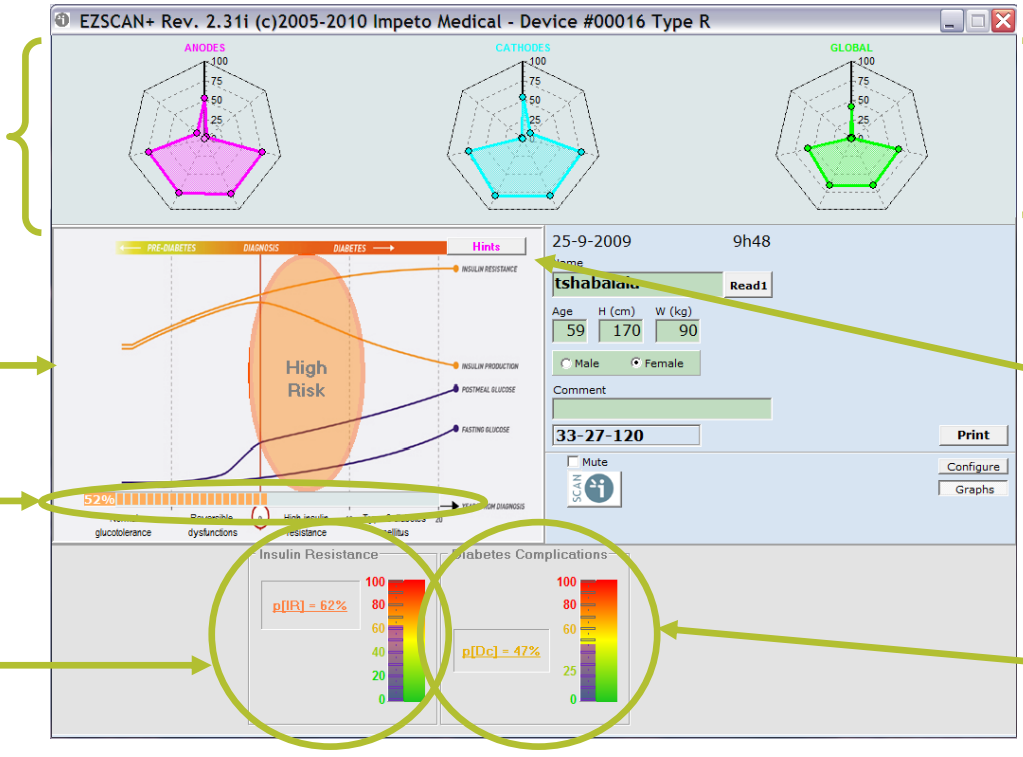
Screening strategy

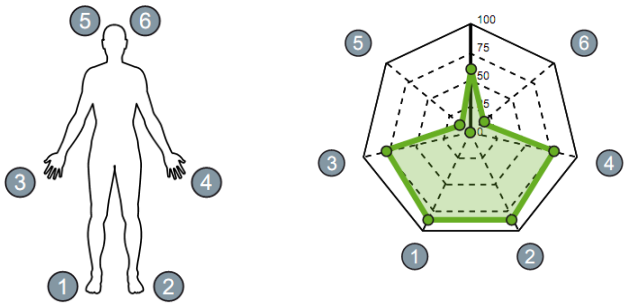
Results are displayed according to diabetes risk with a color index.





Three radars on the top of the screen show the electro conductances measured on each electrode.
A diabetes risk scale is displayed under the color classification.


EZSCAN+ Rev. 2.31i (c)2005-2010 Impeto Medical - Device #00016 Type R
ANODES
CATHODES
GLOBAL
Conductances measured on each electrode
Color classification
Diabetes risk scale
Insulin resistance scale
Hints button
Diabetes complications scale
High Risk
Hints
25-9-2009
9h48
Read1
Age
H (cm)
W (kg)
59
170
90
Male
Female
Comment
33-27-120
Print
Mute
Configure
Graphs
52%
Insulin Resistance
Diabetes Complications
p[IR] = 62%
p[Dc] = 47%


1
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6

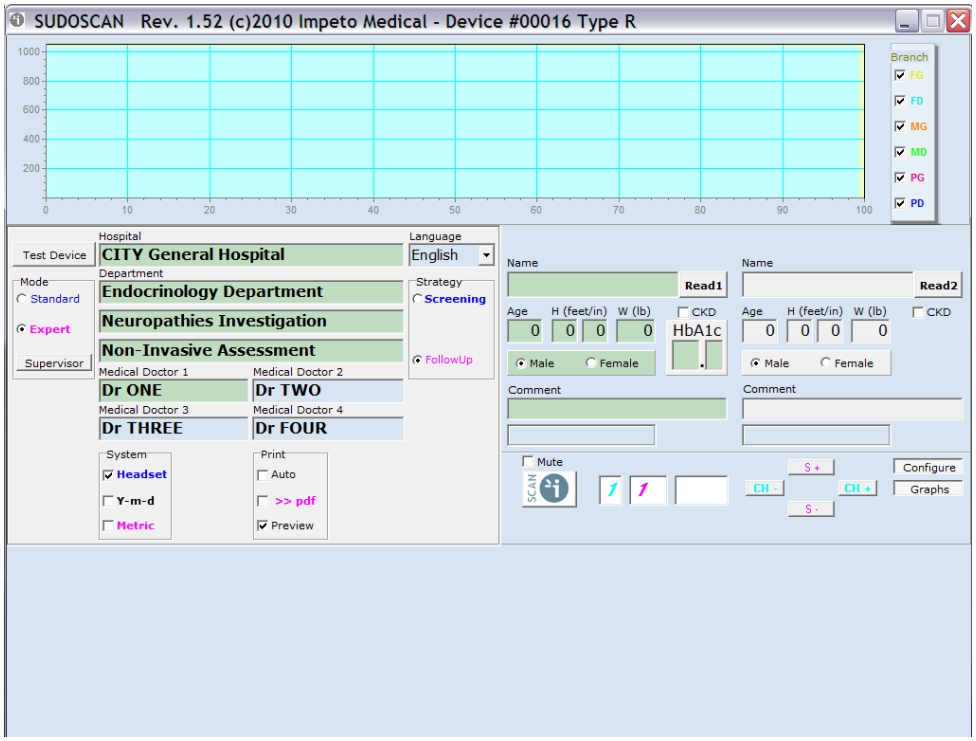
5. Expert Mode

Proceed as described in 4. to run a test.

The expert mode can be used in the context of a follow-up scan.

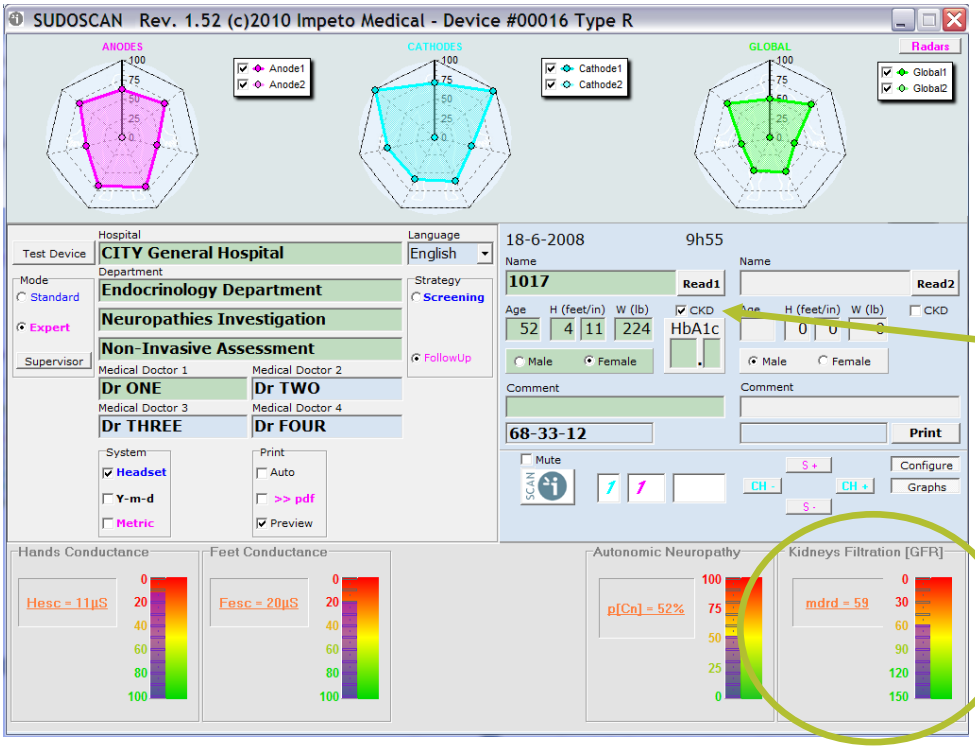
Check the Expert mode box.

Expert checkbox



5.1. CKD scale

In the follow-up strategy the CKD scale can be displayed.



CKD checkbox

Kidney filtration scale

The following instructions are available for both screening and follow-up strategies

5.2. Compare two sets of results

In both strategies, it is possible to compare two results. This function can be useful during a follow up scan. It enables the doctor to assess patient progress.

1. Press the ‘READ 2’ button to open the current folder within the currently selected Medical Doctor.

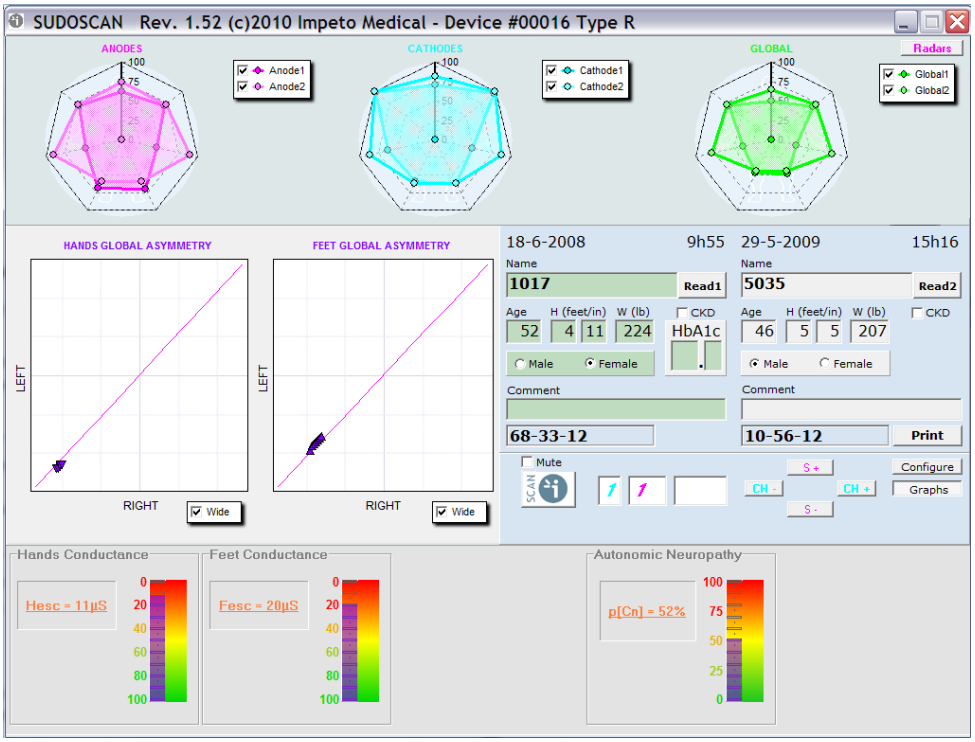
   To select another Medical Doctor (MD1, MD2, etc), click on the ‘up’ button  and select the desired MD.

2. Double-Click on the patient’s name you want to retrieve the data.

NOTE

You can switch from patient 1 data to patient 2 data, by simply touching the corresponding name field and the diagnosis will be updated accordingly for each.

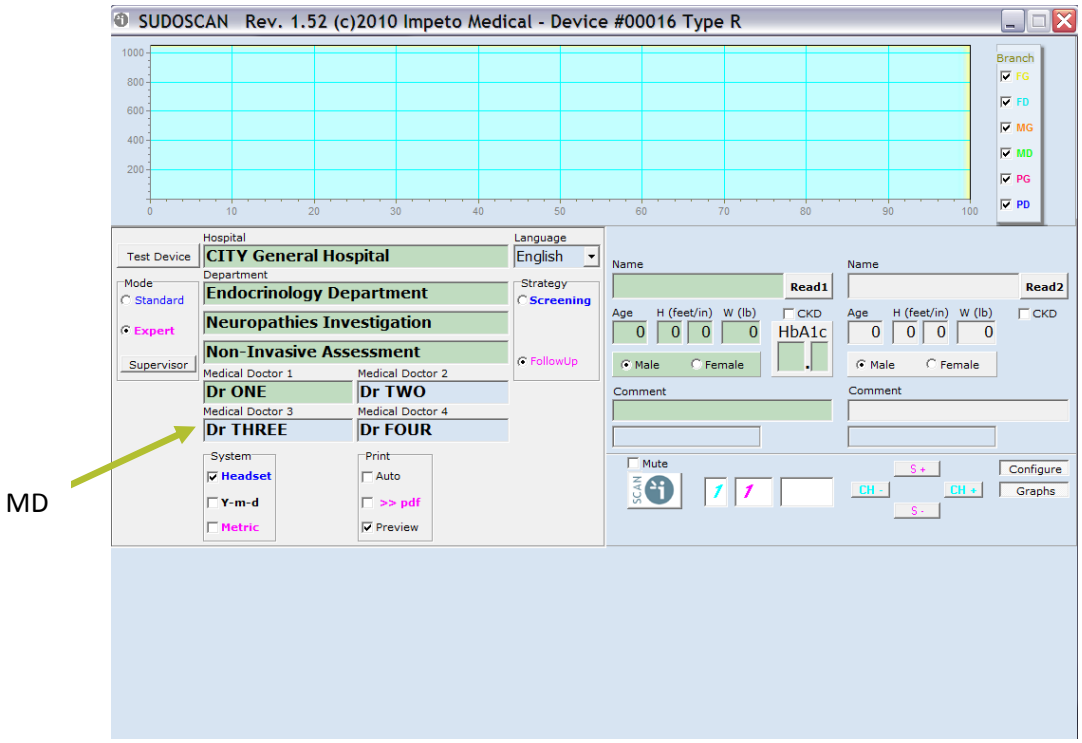
Go to section 7. for more instruction for retrieving patients data.



5.3. Medical Doctor choice

Up to 4 Medical Doctors (MD) can be chosen. The doctor selected last remains selected until later changed (green background). Simply pointing at another doctor's name will select a specific folder for their patients.

Click on the MD you wish. *The background of the selected MD becomes green.*



6. Print the results

The following instructions are available for both standard and expert mode.

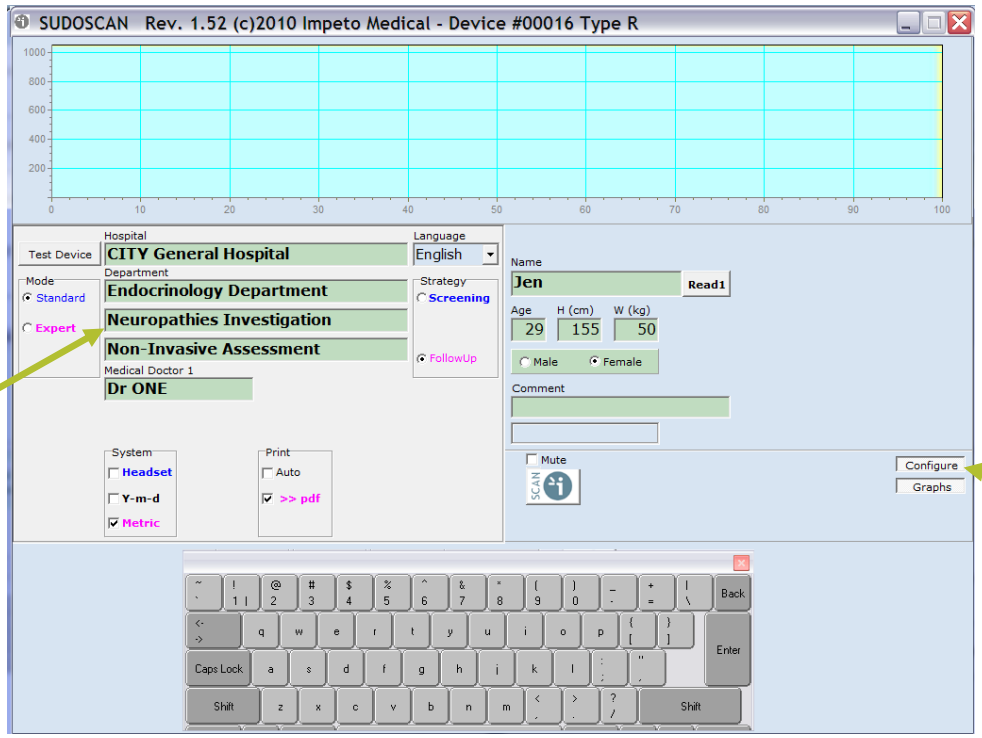
6.1. Configure the print report

1. Press the “Configure” button to open the Configuration window where the print report details can be specified.

2. Double-click on each field to prompt the alphanumerical keyboard to appear on screen.

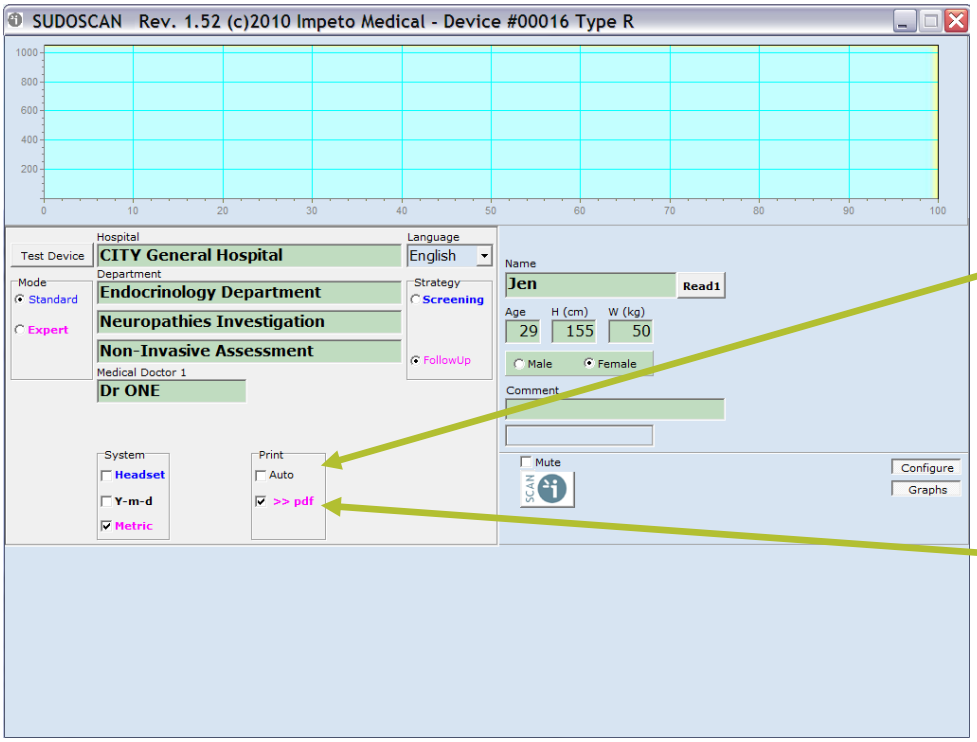
3. Enter the relevant information into the fields:

   NAME OF THE HOSPITAL
   NAME OF THE DEPARTMENT
   NAME OF THE DOCTOR



6.2. Configure the printer usage

Two sets of printing parameters can be selected by the user.



6.2.1. Print auto

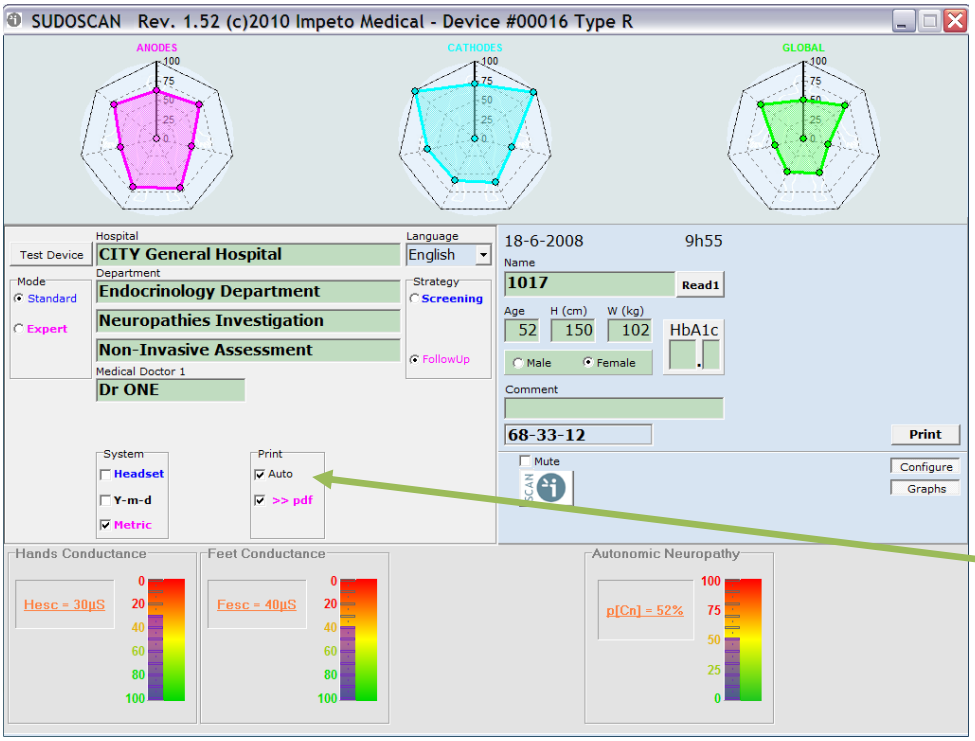
Check the 'Auto' button check-box for automatically printing a test report after the scan completion…

NOTE

Be sure of the printer installation and its configuration settings before selecting the "Auto" mode.

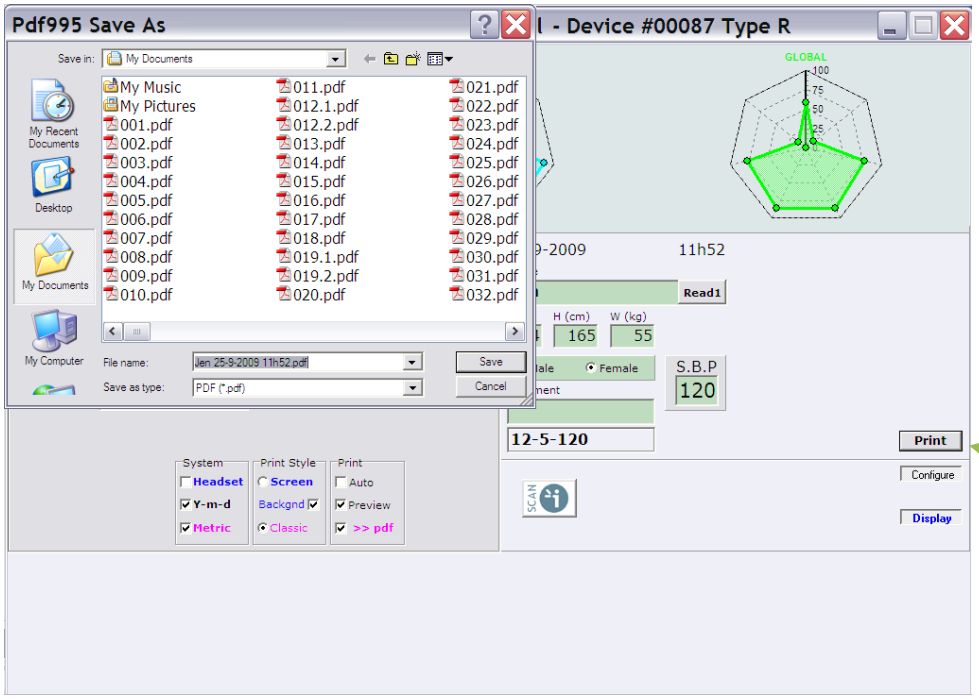
OR

Uncheck the 'Auto' button check-box for selectively manually printing a report only upon pressing the 'PRINT' button.



6.2.2. PDF report

1	Check the ‘pdf’ option check-box.
2	Click on the Print button.
3	By default, the file name is automatically filled with “Name of the patient Date Time” but you can modify it if you want.
4	Save the pdf.



Print button

Press the “Configure” button again to close the Configuration window.

6.2.3. SUDOSCAN report

CITY General Hospital

Endocrinology Department

Neuropathies Investigation

Non-Invasive Assessment

SUDOSCAN REPORT

Patient Identification: akker 12-03-1948

Date and Hour of Examination: 12-11-2009 @ 12h31

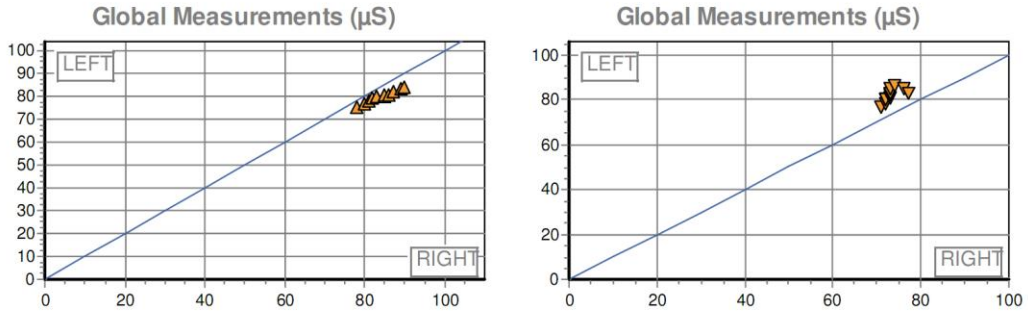
Patient History:

Patient is a 61 year old woman with a BMI of 27

Study Methodology:

Electrochemical Skin Conductances (ESC) of the hands and of the feet are measured using two well-known principles: reverse iontophoresis and electrochemistry. ESC expressed in micro-Siemens (µS) is the ratio between current generated and the constant direct voltage stimulus applied between the electrodes (<4 V). ESC evaluates local sudomotor function. It reflects the lesions of sympathetic nerve fibres that innervate sweat glands. These long and small unmyelinated fibres of the autonomic nervous system are the earliest nerve fibres to undergo damage in peripheral neuropathies (Sumner et al, Neurology 2003). Assessment of symmetry is important to characterize a peripheral neuropathy.

Values of calibration: FEET: 10 kOhms HANDS: 10 kOhms



Conclusion:

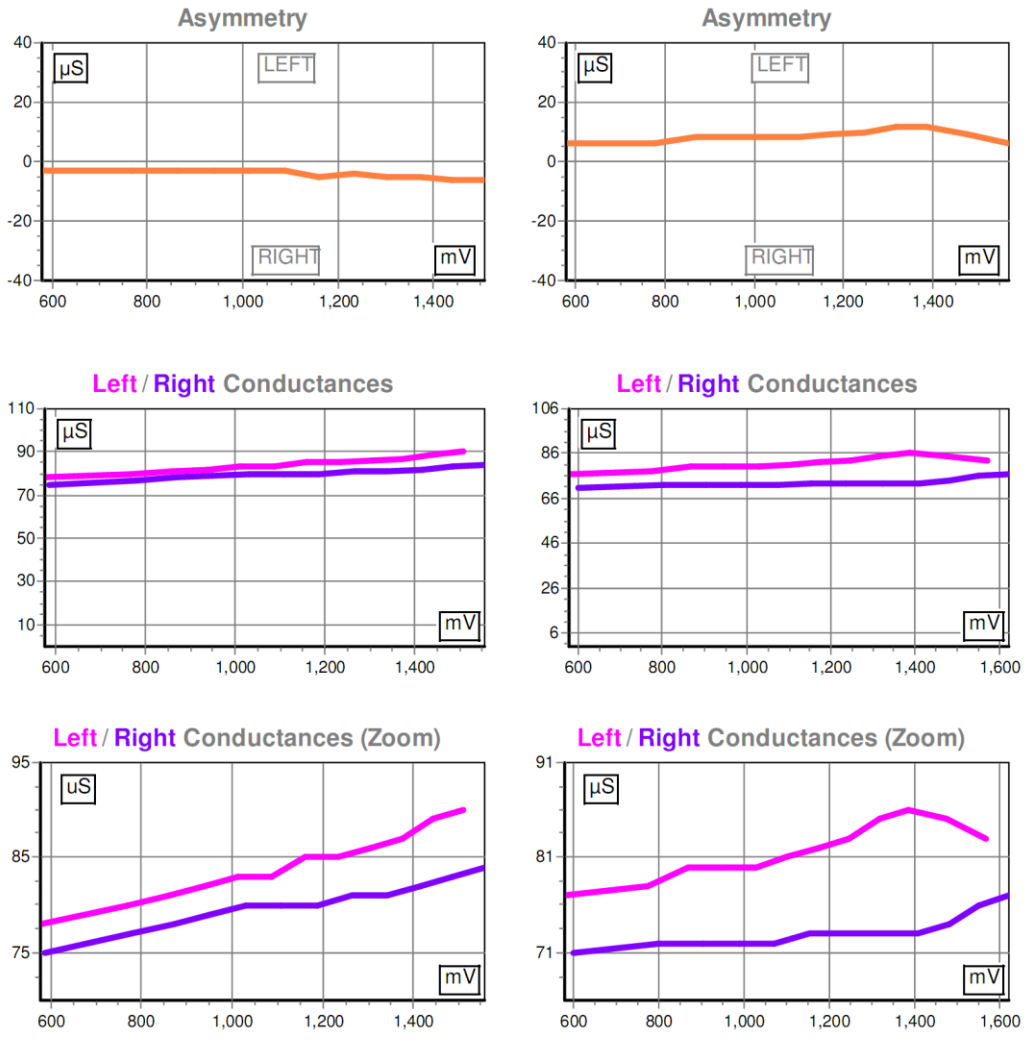
Based on these results, consider absence of peripheral autonomic neuropathy.

No additional exploration is necessary. Plan a new SUDOSCAN test in 1 year time.

Physician comments and signature:

Dr ONE

Feet Measurements: **Hands Measurements:**



Analysis:

Feet Mean ESC: 82 µS (No risk if > 60, Elevated risk if < 40, Moderate risk in between)

Feet Mean Asymmetry: 4 % (No risk if < 20, Possible risk if > 20)

Hands Mean ESC: 77 µS (No risk if > 60, Elevated risk if < 40, Moderate risk in between)

Hands Mean Asymmetry: 10 % (No risk if < 20, Possible risk if > 20)

Risk for cardiac autonomic neuropathy: 33 %

This report provides an objective measurement of sweat function. This assessment of sympathetic nerve function supplements the patient examination; a clinical diagnosis must be made by the physician in the context of all available

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7. Retrieve previous patient data

The following instructions are available for both standard and expert mode.

All patients' data are automatically saved into specific folder on the Master Unit's hard disk. Data are arranged by Medical Doctor (MD1, MD2, etc.), date and time when the scan was done from the most recent to the most ancient.

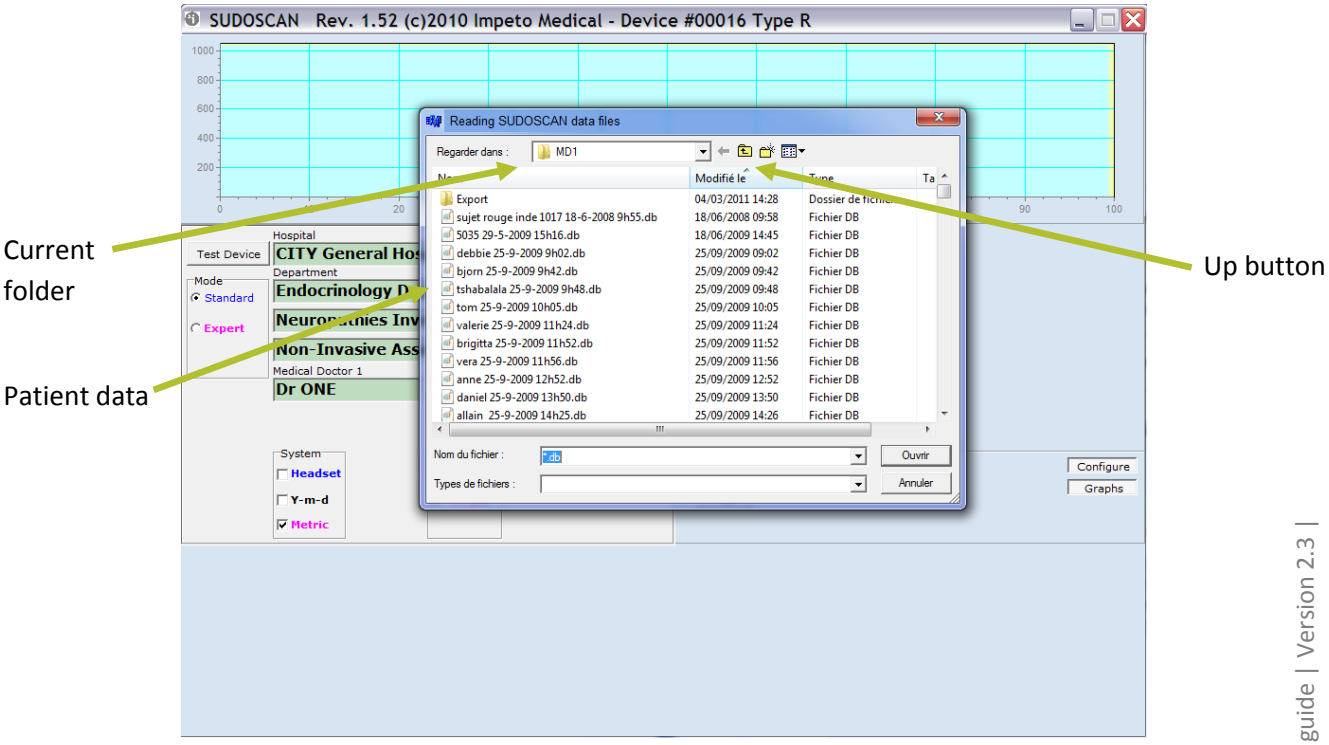
Three methods are available to look for a patient's scan result:

Method 1

1. Double-clicking on the 'Name' field to erase the previous patient's name *(the 'Name' field will become empty).*

2. Press the 'READ 1' button: the whole content of the selected Medical Doctor folder will automatically be displayed in the new window *(all the patients previously scanned).*

3. Double-Click on the patient's name you want to retrieve the data.



Method 2

1. Double-clicking on the 'Name' field to erase the previous patient's name *(the 'Name' field will become empty).*

2. Enter the patient's name you are looking for in the 'Name' field *(double check the spelling)*, then press the 'READ 1' button.

   Only this patient's previous scan results will be displayed in the new window.

Method 3

1. Double-clicking on the 'Name' field to erase the previous patient's name (the 'Name' field will become empty)

2. Enter the patient's first letter(s) name you are looking for in the 'Name' field (double check the spelling), then press the 'READ 1' button.

   All the patients whose names have the same beginning letters will be displayed.

Examples:

Enter 'M' in the 'Name' field and press the 'READ 1' button.

The new window will display all the patients whose names begin with an M.

NOTES

The 'Name' field is NOT case sensitive, and can contain digits (such as Patient ID/IDENTITY CODE).

Since the 'Name' field can contain a maximum of 30 characters, it is advisable to use the 'Comment' field to enter additional patient information such as ID code, etc…

8. Share patient's data

8.1. Transfer data to a USB key

Thanks to this function, the data can be transferred from one SUDOSCAN to another.

All patients' data are saved into a specific database folder on the Master Unit's hard disk, which can be found on C:\Program Files\IMPETO\.

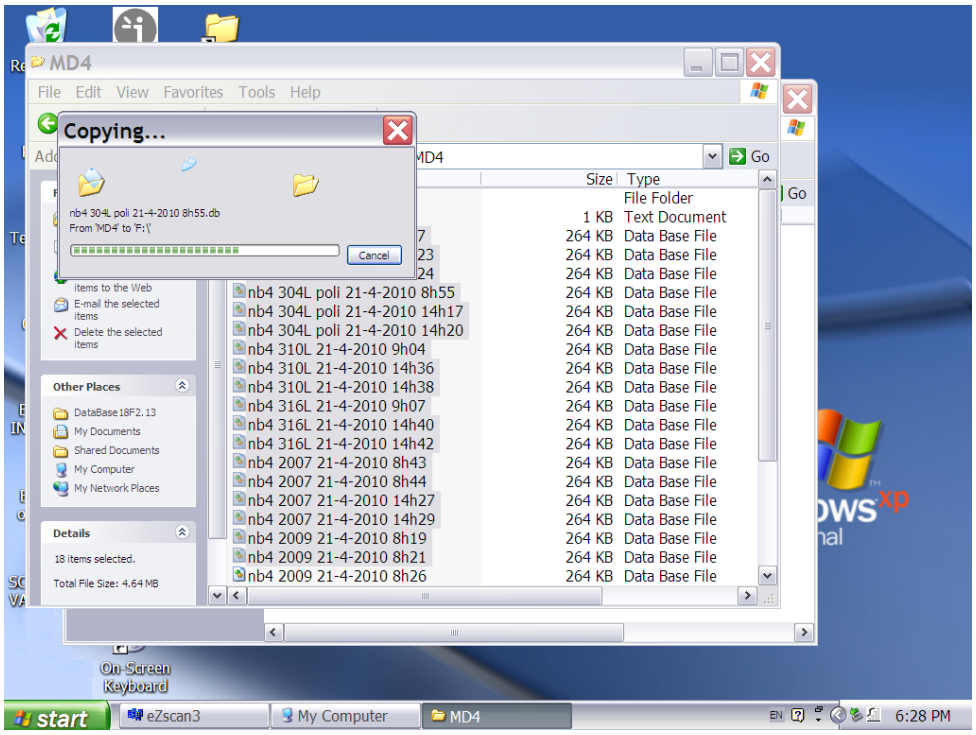
To copy the entire data base:

1	DOUBLE-CHECK ON THE TARGET PC THAT IT IS FREE OF VIRUS.
2	Go to C:\Program Files\IMPETO\
3	Copy the 'DataBase18F2.13' entire folder to the external USB flash disk.



To copy one or more data:

1	DOUBLE-CHECK ON THE TARGET PC THAT IT IS FREE OF VIRUS.
2	Go to C:\Program Files\IMPETO\
3	Click on ‘DataBase18F2.13’
4	Click on MD1, MD2…according to where the data is stored.
5	Copy the Data to the external USB flash disk.



8.2. Access data through office local network

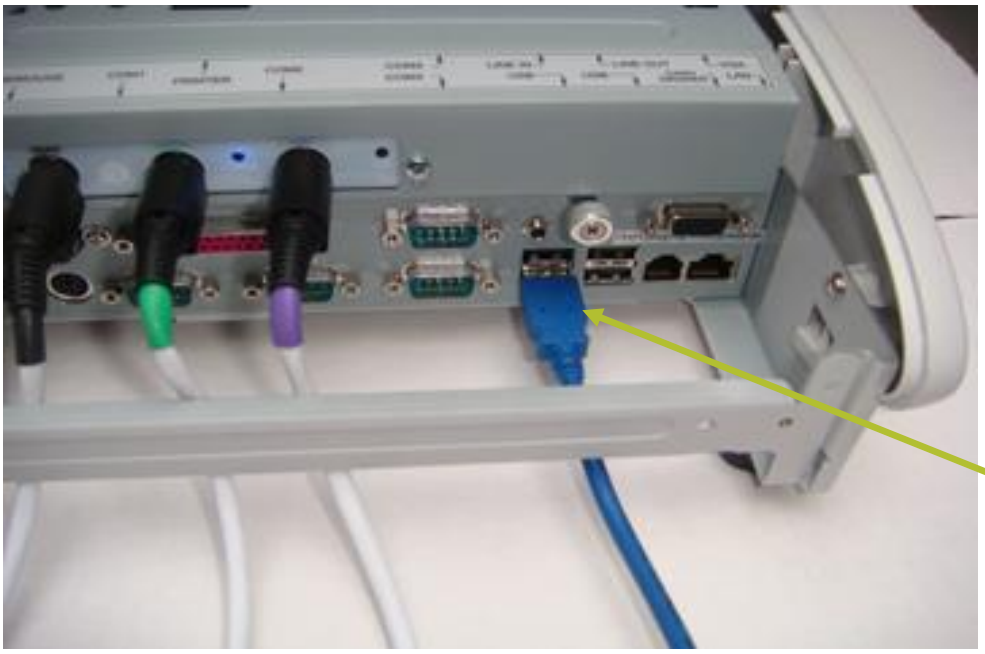
< Requires the installation of SUDOSCAN software on the remote computer >

>>> Check with your distributor.

9. Install a USB printer



1	DOUBLE-CHECKING ON ANOTHER PC THAT THE USB PRINTER IS FREE OF VIRUS.
2	Insert a USB key or a USB-compatible CD-ROM reader with the appropriate printer drivers for Windows XP or Windows 7.
3	Follow the standard Windows procedure to install the new printer drivers and configure the printer default options.



4 Connect the USB printer to one of the 2 USB ports available at the bottom part of the Master Unit's rear panel.

10. Connect a video projector



For a large audience or demonstration purpose, it is possible to connect an external larger screen or a video projector on the VGA-type PC standard video connector located on the right most part of the Master Unit's rear panel.

11. Switch off SUDOSCAN



Always end SUDOSCAN session and turn off the Master Unit only by pressing the on/off button on the front of the Master Unit.

NOTE

The procedure above is causing the device to 'hibernate' so that the same display will be found the next time the Master Unit is powered on.

To fully reset the operation, switch off the machine following standard Windows "turn off" procedure on the display screen, after closing SUDOSCAN application software.

CAUTION

REMOVING THE POWER CORD FROM MASTER UNIT OR FROM WALL SOCKET BEFORE COMPLETE UNIT SHUT OFF, MAY IRREMEDIABLY CORRUPT DATA ON THE UNIT'S HARD DISK.

12. Disconnect the electrodes

Grab the larger part of each connector and slide it softly backwards.

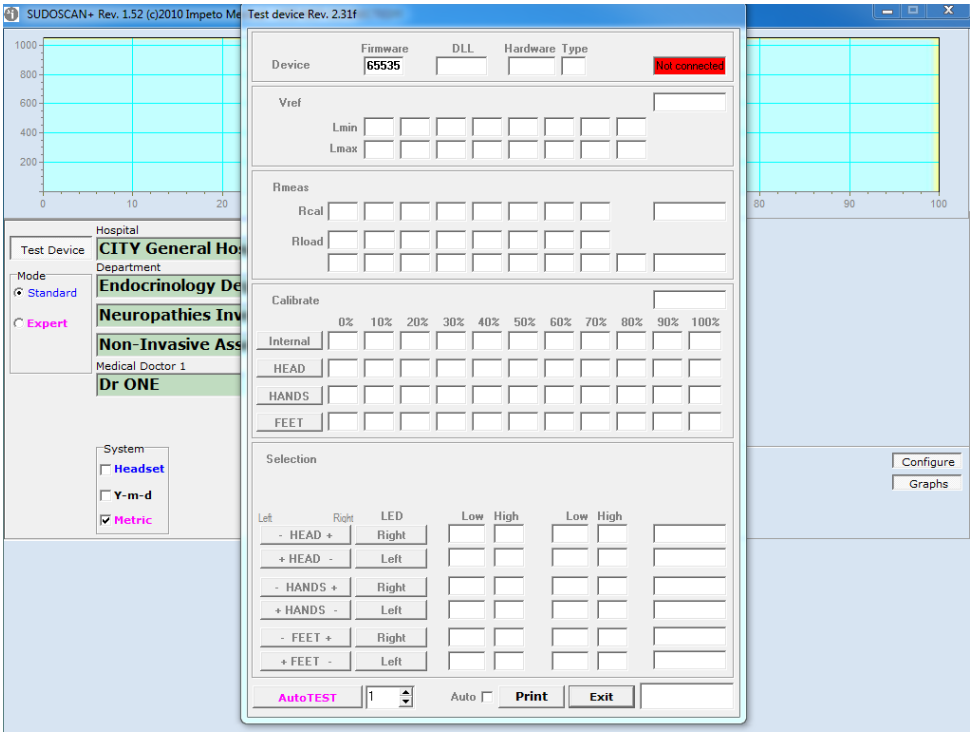




13. Troubleshooting

13.1. Check the cable connections

1. Press the “Configure’ button at the right side of the screen.
2. Press the “Test device” button (need to point to this) at the left side of the new window.



The lower part of this test panel will enable you to check the correct connection of the electrodes:

Pressing either –HANDS+ or +HANDS- buttons, should illuminate both sides of the hands electrodes.

Pressing the “Left” button close to the +HANDS- button should illuminate the left hand electrode only.

Pressing the “Right” button close to the –HANDS+ button should illuminate the right hand electrode only.



Pressing either –FEET+ or +FEET- buttons, should illuminate both sides of the feet electrodes.

Pressing the “Right” button close to the –FEET+ button should illuminate the right foot electrode only.

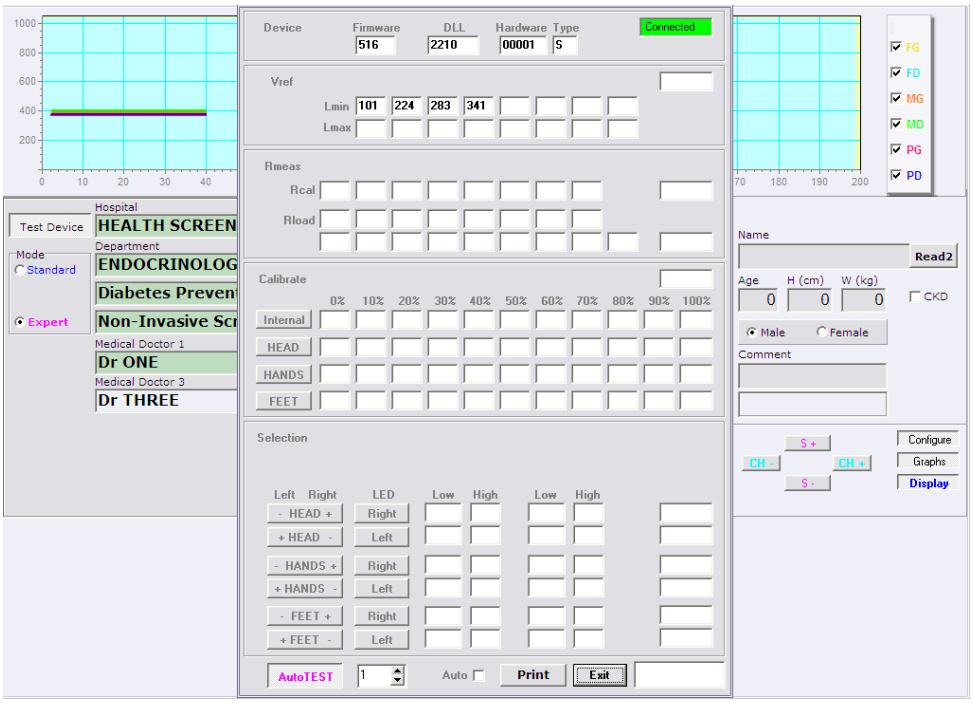
Pressing the “Left” button close to the +FEET- button should illuminate the left foot electrode only.

NOTE

The current aux electrode has no LED attached but should not need to be tested if the hands and feet electrodes are correctly connected to the Master Unit.

Failure to illuminate the correct electrode(s) should prompt you to double-check the physical cable connections and possibly swap the wrong cables before calling for assistance.

13.2. Test the SUDOSCAN hardware



1. Press the ‘Auto Test’ button to initiate the same complete Test and Calibration program which has been launched before leaving the factory; its print report (PDF format) is sitting on the desktop screen.

2. Press the “Exit” button, then the “Configure” button in order to get back to the SUDOSCAN home page.

NOTE

This test must be launched with NO PATIENT installed on electrodes.

13.3. Calibrate the touch screen

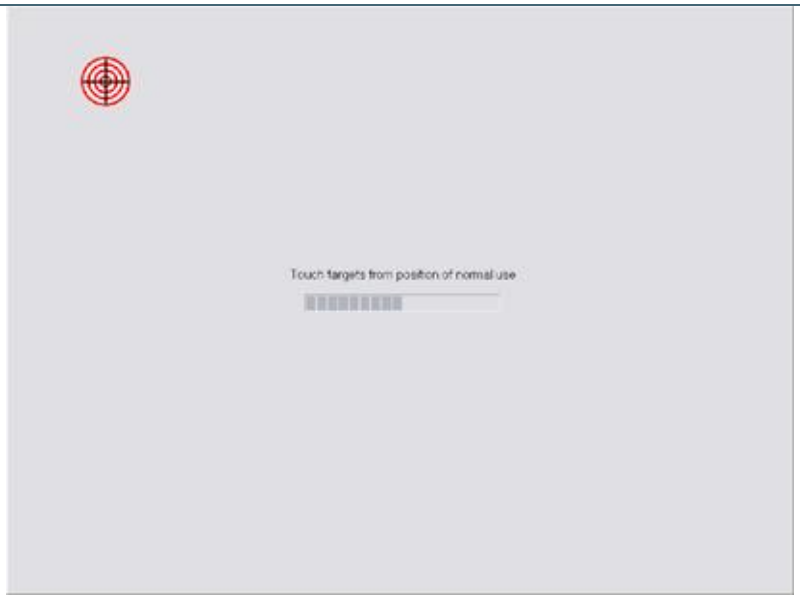
In case the pointing capability on the touch screen does not appear accurate enough, follow the following procedure to calibrate or 'fine-tune' the SUDOSCAN touch screen operation:

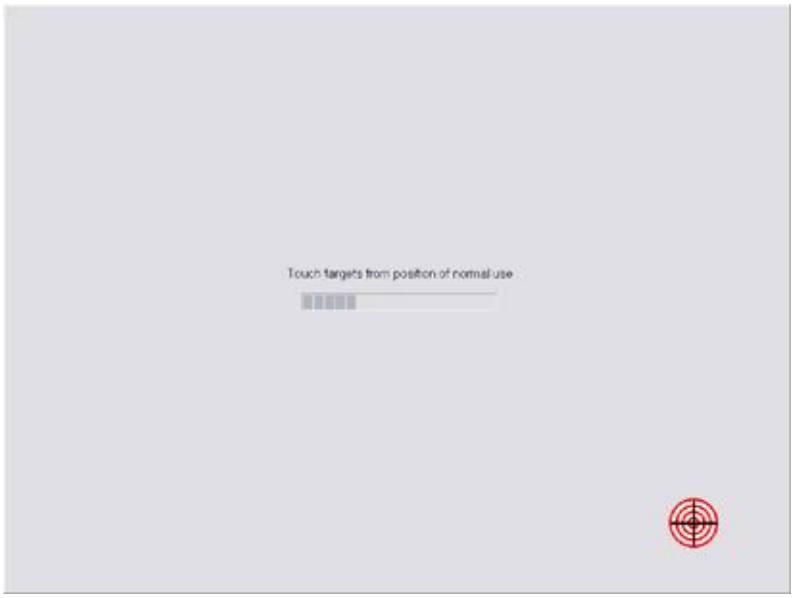
1 Click on the ‘ELO Touch’ icon on the left to launch the ‘Touch Utility’ program.



2 Click on the ‘Align’ icon to launch a simple 3-step procedure.

3 When requested, press with your finger on the 3 targets presented sequentially.


Touch targets from position of normal use


Touch targets from position of normal use

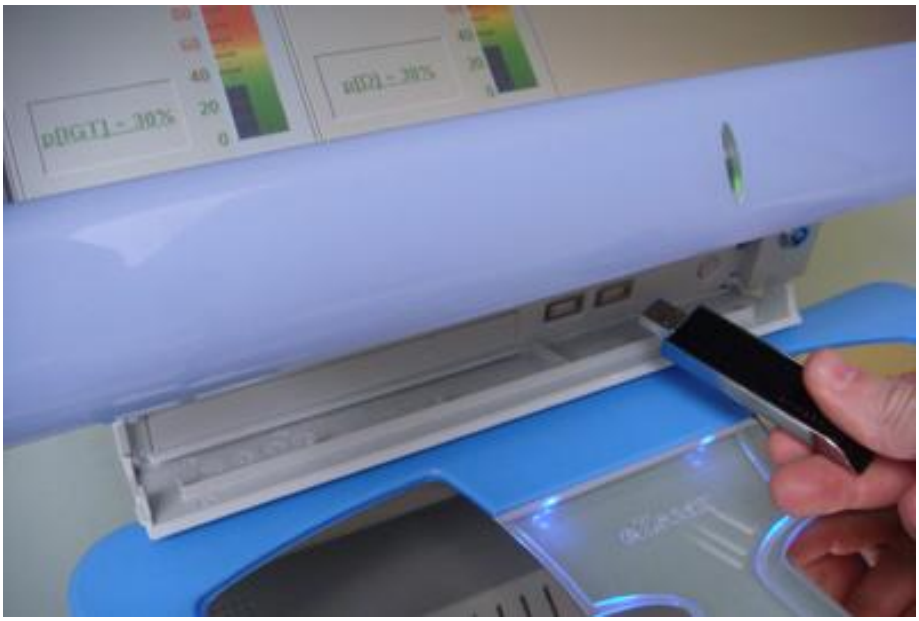

Touch targets from position of normal use

4 Press ‘Confirm’ box.



14. Upgrade SUDOSCAN firmware

Impeto Medical will regularly provide software upgrades to ensure continuous enhancement of SUDOSCAN products. Check with your distributor or on www.impeto-medical.com website for their availability on a regular basis.



1	Close SUDOSCAN program.
2	AFTER DOUBLE-CHECKING ON ANOTHER PC THAT IT IS FREE OF VIRUS, insert a USB Flash Disk into one of the 2 USB ports available on the front panel below the screen.
3	Click on the self-extractible software upgrade file for automatic upgrading.
4	Click on SUDOSCAN icon in the middle of the desktop screen.

15. Precaution of use and technical specifications

Principle of SUDOSCAN

SUDOSCAN® is based on two well-known principles, reverse iontophoresis and electrochemistry. SUDOSCAN collects physiological data by means of Chronoamperometry and processes them with analysis software:

1. Reverse iontophoresis: this technology applies a low DC voltage between electrodes located on specific areas of the skin with high density of appendage ducts in order to evaluate the electroporation level of subdermal concentrations (Chimadzhev *et al.*, 1998),
2. Electrochemistry: this technology applies a sudden step of DC low voltage between two electrodes in a fluid to obtain data related to the concentration of this solution.

SUDOSCAN® devices use both technologies, coupled and complemented with exclusive software which analyzes cutaneous conductance data collected through the Chronoamperometry measurements.

The SUDOSCAN system is composed of a SUDOSCAN electronic device integrated into a touch-screen Windows XP-based Industrial-grade PC computer and connected to 6 electrodes placed on the feet, the hands and head, as well as SUDOSCAN acquisition control software also responsible for archiving the measurements taken on all patients.

General operation of SUDOSCAN

The patient positions his bare feet on the feet electrodes, applies the palms of his hands to the hands electrodes, and the operator can affix forehead electrodes to the head.

After entering the information specific to the patient (name, age, gender, height, weight), the operator will ask the software to activate the electronic circuitry of the Chronoamperometry device, which will apply DC reference voltages successively to electrode couples and will measure the evolution in time of the electrical current passing through the electrodes.

Several successive cycles of measurements are carried out in an automatic way and all the measured values are recorded on the hard disk. Data-processing is then performed to compute the conductance on each electrode.

The number of measurements made by means of the SUDOSCAN device, as well as the number of the recorded patients is only limited by the capacity of the integrated hard disk.

At the end of the measurements cycle, which lasts approximately 2 minutes, the expert can visualize a synthesis of the reprocessed data of the patient and with the possibility of printing a sheet of results to include in the file of the patient.
No control is accessible to the patient and the use of the SUDOSCAN system is strictly reserved to physicians in the framework of a professional use in medical clinic or during visits in hospital.

Precautions of use and maintenance

Transport

The system SUDOSCAN is delivered under the conditions of packing described previously.

If it is necessary to pack, to transport or deliver SUDOSCAN after its use, it is recommended to arrange all its elements in their housing and position of origin.

Pay particular attention to carefully arrange the electrode cables in the locations especially designed for them in the protection foams, not to damage them or weaken them.

The SUDOSCAN box can be stored upright or laid down.
The SUDOSCAN device does not comprise any accumulator.

Recycling

At the end of its lifetime, the SUDOSCAN system must be returned to Impeto Medical's authorized distributor, which will return it to Impeto Medical, in order to ensure the recycling of certain components.

The components and the accessories of SUDOSCAN are free of mercury and of components containing this element.

Protection against the very wet environments

It is advised not to use SUDOSCAN in a very wet environment.

SUDOSCAN is not protected from the effects of the discharges of an external defibrillator, nor against high frequency currents or strong electromagnetic disturbances. The use of portable telephones or wireless fixed telephones can cause disturbances of the signal.

Electromagnetic compatibility

The use of portable and mobile RF communication devices (for example: cellular telephones) can influence the analysis carried out during the recording, as the recorded signals can be disturbed by electromagnetic interferences.

The device should not be used in the presence of ionizing radiations (x-rays, gamma rays …) because those could erase the internal storage.
For the tables concerning the electromagnetic emissions and the immunity of the recorder, see the appendices of this handbook.

Maintenance of the device and electrodes

No particular maintenance is necessary.

The external case and the cables of the electrodes can be cleaned using a slightly wet tissue or soaked with soapy water. Do NOT use any detergent product, nor alcohol or acetone.

To avoid corrosion like ensuring the best conditions of hygiene, the feet and hands electrodes must be cleaned immediately after each patient with an anti-bacterial and fungicidal solution.

Warranty

SUDOSCAN has one full year warranty period (12 months from the date of installation, parts and labor). After the warranty period has expired, customers can subscribe a service contract for the maintenance of their equipment.

Feet and hands electrodes, as well as their associated cables will have their wear and tear and hence are not covered by the warranty. Take care to maintain them in good condition of aspect and operation to prolong their lifespan to the maximum, then to order the replacement parts at Impeto Medical or its authorized distributors.

WARNING! The warranty is null and void if SUDOSCAN was opened and repaired by any unauthorized person. Warranty will hold well only against manufacturing defects and certainly not for any mechanical damage due to mishandlings. Refer to manual for proper use of the equipment.

Description of the pictograms affixed on the case

The following pictograms are affixed on SUDOSCAN:



Indicates that the parts applied of the device are of type BF (IEC 60601-1).

III

Class of electric protection.

CE 0499 SUDOSCAN is in conformity with European Directive 93/42/EEC.

Use of the electrodes

The data relating to the use of the electrodes are provided as an indication.

To ensure high quality recordings, a particular care must be taken to the preparation of the skin of the patient and to the installation of the electrodes.

It is also advisable to check if the electrode cable connectors are well inserted in the corresponding sockets at the rear of the SUDOSCAN Master Unit.

Clean the skin of the patient to establish a good electrode/skin contact.

WARNING! Use only the original electrodes provided by Impeto Medical.

Impeto Medical cannot guarantee the results of measurements carried out with electrodes coming from other manufacturers, even if they have a similar appearance!

When it is not used, the SUDOSCAN system can be stored without being necessary to disconnect the electrode cables.

Forehead electrodes installation

Put in place the head gear with the reusable electrodes and adjust its size to the patient head. The blue side of the cable will be connected to the electrode placed on the right side of the patient's head the white side on the left.

Positioning on the electrodes plates

The patient must position the sole of his bare feet in the center of the feet electrodes and his hands well flat in the center of the hands electrodes, hands must be applied firmly. Except in the event of characterized overweight, the patient will be preferably standing upright on the plate of the feet electrodes.

Interrupting or stopping the recording

The recorder will stop automatically at the end of the measurements cycle, after approximately 2 minutes.

To stop the recorder BEFORE the end of the measurements cycle, simply click on the same button which is now labeled "STOP" during the cycle of measurement.

Recommendations to the operator

As for any medical examination, it is recommended to check the environmental conditions and the condition of the patient before carrying out a measurement:

- Room temperature between 18 and 25 degrees Celsius,
- Patient not having carried out an intense effort during 2 hours preceding the examination (except in the event of effort test),
- Cardiac holter measurements going on can be altered during the 2 minutes time scan.

Contra-indications

The repeated use of SUDOSCAN does not create any side effect, but no specific tests have been carried out to date under the following particular conditions, so it is advised not to proceed to measurements on the following patients:

- People with any open or bleeding wound at any electrode contact surface location,
- People with any type of implantable device,
- People with missing leg(s) or hand(s),
- Pregnant Women or women who are uncertain about a possible pregnancy, Subjects under the age of 21.

Calibration

SUDOSCAN performs automatically a self-calibration procedure before each patient. So it assures that each electrode got enough contacts to proceed for the measurements. Therefore, there is no need to calibrate the device before operation. However, it is possible to double-check the correct operation of the elements of the whole system by carrying out, in case of doubt or at least once in a year a calibration procedure using the “Calibration Test Box” accessory available from Impeto Medical.

Accessories and disposable parts

The following elements are available for SUDOSCAN:

Code	**Description**
SuMU	SUDOSCAN Master Unit
APH2	Hands plate with two electrodes and its connecting cable
APF2	Feet plate with two electrodes and its connecting cable
HDGR	Head gear with two electrodes
SCC2	Head electrodes cable with 2 clips
UMSu	Installation manual
ASAFE	Bottle of ANIOS Surfa’Safe spray cleaner

Technical specifications

• Description

SUDOSCAN is a device capable of measuring galvanic skin responses and is designed for the medical professional.

SUDOSCAN is a class IIa medical device (according to the EEC directive 93/42) and a class II medical device (according FDA 21CFR 882.1540).

• Characteristics of acquisition

Storage support:	80 Gb Hard disk
Mode:	No compression
Frequency of acquisition:	100 Hz
Resolution:	1 nanoSiemens
Display units:	microSiemens
Display precision:	+ or – 1%

• Electrical characteristics

Measurement voltage:	1V to 4V DC typical
Frequency:	0 Hz (DC current)
Dynamic resolution:	10 bits
Power requirement of the unit:	100 - 240V AC + or – 3% variation at 50 - 60Hz freq
Power supply:	250 W power supply in conformity with the EN60601-1

• Sealing

Index of protection:	IP41

• Mechanical characteristics

Master Unit alone	13 kg
Master Unit dimensions	38 x 38 x 30 cm
Weight of the Master Unit + electrodes:	15 kg
Weight of the transportation box:	16 kg
Dimensions of the shipping box:	54 x 46 x 39 cm

Storage and conditions of use of the apparatus and electrodes

	Temperature	Moisture	Pressure (without condensation)
Operation	15° to 35°C	20% to 80%	700 to 1060 hPa
Storage and transport	-0° to 45°C	20% to 80%	500 to 1060 hPa

Table 1

Guidance and manufacturer's declaration – electromagnetic emissions		
SUDOSCAN is intended for use in the electromagnetic environment specified below. The customer or the user of SUDOSCAN should assure that it is used in such an electromagnetic environment.		
Emissions test	**Compliance**	**Electromagnetic environment – guidance**
RF emissions CISPR 11	Group 1	SUDOSCAN uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class B	SUDOSCAN is suitable for use in all establishments, including domestic establishments and those directly connected to the public low voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC 61000-3-2	Class A	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Complies	

Table 2

Guidance and manufacturer's declaration – electromagnetic immunity			
SUDOSCAN is intended for use in the electromagnetic environment specified below. The customer or the user of the SUDOSCAN should assure that it is used in such an electromagnetic environment.			
Immunity test	**IEC 60601 test level**	**Compliance level**	**Electromagnetic environment guidance**
Electrostatic discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	±6 kV contact ±8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30 %.
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	±2 kV for power supply lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV line(s) to line(s) ±2 kV line(s) to earth	±1 kV mode differential ±2 kV mode common	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5 % *UT* (>95 % dip in *UT*) for 0,5 cycle 40 % *UT* (60 % dip in *UT*) for 5 cycles 70 % *UT* (30 % dip in *UT*) for 25 cycles <5 % *UT* (>95 % dip in *UT*) for 5 s	0 Vac during 10 ms 92 Vac during 100 ms 161 Vac during 1 s 0 Vac during 5 s	Mains power quality should be that of a typical commercial or hospital environment. If the user of the SUDOSCAN requires continued operation during power mains interruptions, it is recommended that the SUDOSCAN be powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m for 50Hz and for 60Hz	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
NOTE *UT* is the a.c. mains voltage prior to application of the test level			

Table 3 - for ME equipment and ME systems that are not life-supporting

Guidance and manufacturer's declaration – electromagnetic immunity			
The SUDOSCAN is intended for use in the electromagnetic environment specified below. The customer or the user of the SUDOSCAN should assure that it is used in such an electromagnetic environment.			
Immunity test	**Immunity test**	**Immunity test**	**Immunity test**
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 V rms 3 V/m	Portable and mobile RF communications equipment should be used no closer to any part of the SUDOSCAN, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. **Recommended separation distance** $d = 1.17 \sqrt{P}$
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2,5 GHz		$d = 1.17 \sqrt{P}$ 80 MHz to 800 MHz $d = 2.33 \sqrt{P}$ 800 MHz to 2,5 GHz where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey (1) should be less than the compliance level in each frequency range. (2) Interference may occur in the vicinity of equipment marked with the following symbol: 
NOTE 1 At 80 MHz and 800 MHz, the higher frequency range applies. NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

(1) Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the SUDOSCAN is used exceeds the applicable RF compliance level above, the SUDOSCAN should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the SUDOSCAN.

(2) Over the frequency range 150 kHz to 80 MHz, field strengths should be less than [*V*1] V/m.

Table 4 – for ME equipment and ME systems that are not life-supporting

Recommended separation distances between portable and mobile RF communications equipment and the SUDOSCAN			
SUDOSCAN is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the SUDOSCAN can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the SUDOSCAN as recommended below, according to the maximum output power of the communications equipment.			
Rated maximum output power of transmitter W	**Separation distance according to frequency of transmitter** m		
	150 kHz to 80 MHz $d = 1.17 \sqrt{P}$	**80 MHz to 800 MHz** $d = 1.17 \sqrt{P}$	**800 MHz to 2,5 GHz** $d = 2.33 \sqrt{P}$
0,01	0.12	0.12	0.23
0,1	0.37	0.37	0.74
1	1.16	1.16	2.33
10	3.69	3.69	7.37
100	11.6	11.6	23.3
For transmitters rated at a maximum output power not listed above, the recommended separation distance *d* in metres (m) can be determined using the equation applicable to the frequency of the transmitter, where *P* is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer. NOTE 1 At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies. NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

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DEPARTMENT OF HEALTH & HUMAN SERVICES

Public Health Service

Food and Drug Administration
10903 New Hampshire Avenue
Document Control Room -WO66-G609
Silver Spring, MD 20993-0002

Impeto Medical
c/o Mr. Kye Cheung
4479 Shorepointe Way
San Diego, CA 92130

JUN 14 2010

Re: K100233
Trade/Device Name: Impeto Medical SudoScan
Regulation Number: 21 CFR 882.1540
Regulation Name: Galvanic skin response measurement device
Regulatory Class: Class II
Product Code: GZO
Dated: June 4, 2010
Received: June 4, 2010

Dear Mr. Cheung:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.



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